A Project Report

On

MEDICAL TEXT CLASSIFICATION SYSTEM FOR CATEGORIZING HEALTH QUERIES USING NLP

Submitted to

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY ANANTHAPUR, ANANTHAPURAMU

In Partial Fulfillment of the Requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND TECHNOLOGY

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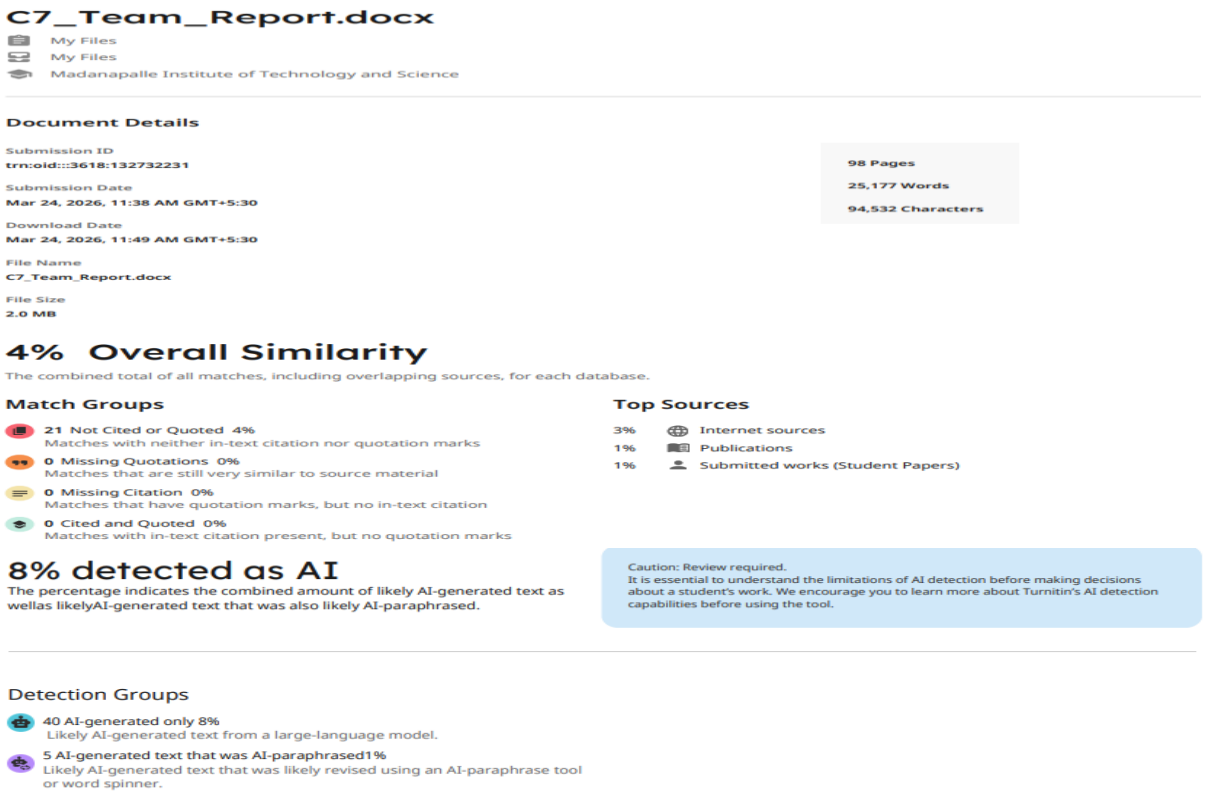
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ABSTRACT

The increasing volume of online health-related queries has made it challenging for healthcare platforms and medical support systems to provide quick and organized responses. Users often ask questions in informal language, making it difficult to categorize and process them manually. This System aims to develop an intelligent Medical Text Classification System using Natural Language Processing (NLP) and deep learning techniques to automatically categorize health queries into meaningful classes such as symptoms, diseases, medications, diagnosis, and treatment options. The system preprocesses patient queries, extracts semantic features, and classifies them using models like LSTM, Bi-LSTM, or transformer-based architectures such as BERT. By automating the classification process, the proposed system enables faster response times, reduces manual workload, and improves the accuracy and efficiency of online healthcare assistance. This approach benefits digital health platforms, telemedicine services, and medical chatbots by helping users access the right information more quickly and effectively.

Keywords: *Medical Text Classification, Natural Language Processing (NLP), Deep Learning, LSTM, Bi-LSTM, BERT, Healthcare Analytics, Text Preprocessing, Semantic Feature Extraction, Health Query Classification, Telemedicine, Medical Chatbots.*

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LIST OF ABBREVIATION

NLP	Natural Language Processing
ML	Machine Learning
LSTM	Long Short-Term Memory
BiLSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
GRU	Gated Recurrent Unit
RNN	Recurrent Neural Network
BERT	Bidirectional Encoder Representations from Transformers
NLTK	Natural Language Toolkit
TF-IDF	Term Frequency — Inverse Document Frequency
OOV	Out-of-Vocabulary
API	Application Programming Interface
DFD	Data Flow Diagram
UML	Unified Modeling Language
GPU	Graphics Processing Unit
CPU	Central Processing Unit
NLP	Natural Language Processing
ML	Machine Learning

CHAPTER -1
INTRODUCTION

1.1 Motivation

The rapid growth of digital healthcare platforms, telemedicine services, and online medical forums has resulted in a significant increase in health-related queries submitted by users. Patients frequently post questions about symptoms, diseases, medications, and treatments through websites, mobile applications, and chat-based systems. However, these queries are often written in informal language, contain spelling errors, or lack proper medical terminology. As a result, manual categorization becomes inefficient, time-consuming, and inconsistent.

Healthcare support systems require an intelligent mechanism to automatically organize and classify such queries into meaningful medical categories. Traditional keyword-based approaches fail to capture contextual meaning and semantic relationships between medical terms. Therefore, there is a strong need for an advanced Natural Language Processing (NLP)-based solution that can understand user intent and classify medical queries accurately.

The primary motivation behind this system is to design and develop an intelligent Medical Text Classification System that can automatically categorize health-related queries using deep learning models such as LSTM, Bi-LSTM, and transformer-based architectures like BERT. By leveraging contextual embeddings and semantic feature extraction, the system aims to enhance the efficiency, speed, and reliability of digital healthcare assistance platforms.

The key objectives driving this work include:

1. **Automating Medical Query Categorization** – Reducing dependency on manual sorting of health-related queries.
2. **Improving Response Efficiency** – Enabling faster routing of queries to appropriate healthcare resources.
3. **Enhancing Classification Accuracy** – Utilizing deep learning models to capture contextual and semantic meaning.
4. **Supporting Digital Healthcare Systems** – Assisting telemedicine platforms and medical chatbots in delivering structured and reliable responses.

   By integrating NLP with deep learning, this system contributes to improving intelligent healthcare communication systems.

.

1.2 Problem Definition

With the increasing use of online healthcare services, thousands of medical queries are generated daily. These queries are typically unstructured, written in natural conversational language, and vary significantly in terminology and format. Manual classification of such queries into categories such as symptoms, diseases, medications, diagnosis, and treatment options is inefficient and prone to human error.

Traditional text classification approaches rely on keyword matching or rule-based systems, which fail to understand contextual meaning and medical semantics. Additionally, variations in patient language, spelling mistakes, abbreviations, and incomplete sentences further complicate accurate categorization.

The primary problem addressed in this system is the development of an automated system capable of accurately classifying unstructured medical queries into predefined categories using Natural Language Processing techniques.

Key challenges include:

- **Unstructured Text Data** – Queries are written in informal, inconsistent formats.
- **Semantic Ambiguity** – The same medical term may have different meanings depending on context.
- **Imbalanced Data Distribution** – Certain categories may contain significantly more data than others.
- **Domain-Specific Terminology** – Medical language requires contextual understanding beyond simple keyword matching.
- This system aims to overcome these challenges by implementing deep learning-based classification models capable of understanding contextual relationships within medical text.

1.3 Objective of the System

The primary objective of this system is to design and implement an intelligent Medical Text Classification System using Natural Language Processing and deep learning techniques.

The specific objectives are:

- To preprocess and clean unstructured medical query data.
- To extract meaningful semantic features using embedding techniques such as TF-IDF, Word2Vec, or BERT embeddings.

- To implement and compare deep learning models such as LSTM, Bi-LSTM, and transformer-based architectures.
- To classify medical queries into predefined categories such as symptoms, diseases, medications, diagnosis, and treatment.
- To evaluate model performance using metrics such as accuracy, precision, recall, and F1-score.
- To develop a scalable system that can be integrated into healthcare platforms and medical chatbots.

1.4 Scope of the System

The scope of this system focuses on the automated classification of health-related textual queries using NLP and deep learning models. The system is designed to handle English-language medical queries collected from online healthcare forums, telemedicine platforms, and public medical datasets.

The system includes:

- Text preprocessing and normalization.
- Feature extraction using word embeddings.
- Implementation of LSTM, Bi-LSTM, and BERT-based models.
- Performance evaluation and comparison of models.
- Integration capability for real-time medical support systems.
- The system does not aim to provide medical diagnosis or clinical decision-making but instead focuses on organizing and categorizing user queries to improve response management in digital healthcare platforms.

CHAPTER – 2

LITERATURE SURVEY

2.1 Overview

Medical text classification sits at the intersection of healthcare and **Natural Language Processing (NLP)**. The basic task is to take a piece of text—such as a patient query, clinical note, discharge summary, or forum post—and assign it to one or more meaningful labels (for example: symptoms, diagnosis, medication-related, etc.). In online healthcare systems, the text is often short and conversational, which makes the classification problem slightly different from clinical-document classification.

People frequently use incomplete sentences, local expressions, misspellings, and mixed medical/non-medical terms. This is exactly why a strong literature base is important: earlier research provides multiple approaches (rule-based systems, classical machine learning, deep learning, and transformer methods), each with different strengths and weaknesses. The discussion in this chapter builds a clear path from older text categorization techniques to modern deep-learning-based medical classification approaches.

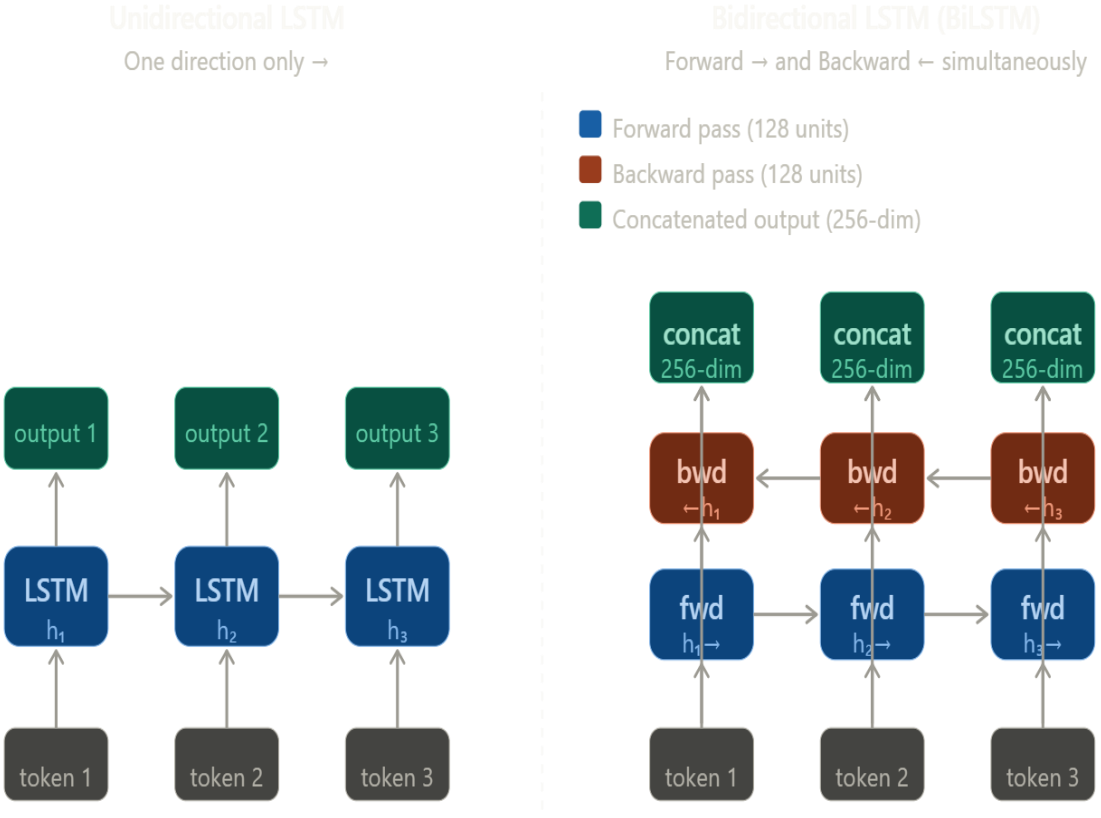


Fig 2.1 Unidirectional LSTM vs Bidirectional LSTM (BiLSTM) Architecture

2.2 Related Work

Early medical query routing systems often relied on hand-crafted rules and keyword matching. For example, if a query contained “fever” or “headache” it was labelled as symptoms, and if it contained “tablet” or “side effect” it was labelled as medications. This approach is simple to implement and easy to interpret. However, such systems fail when the wording changes, and they struggle to disambiguate intent.Classical machine learning approaches treat text classification as a supervised learning problem using methods such as Naïve Bayes, Logistic Regression, SVM, or Random Forest. These methods became popular because they were more flexible than pure rules and could learn patterns from data.

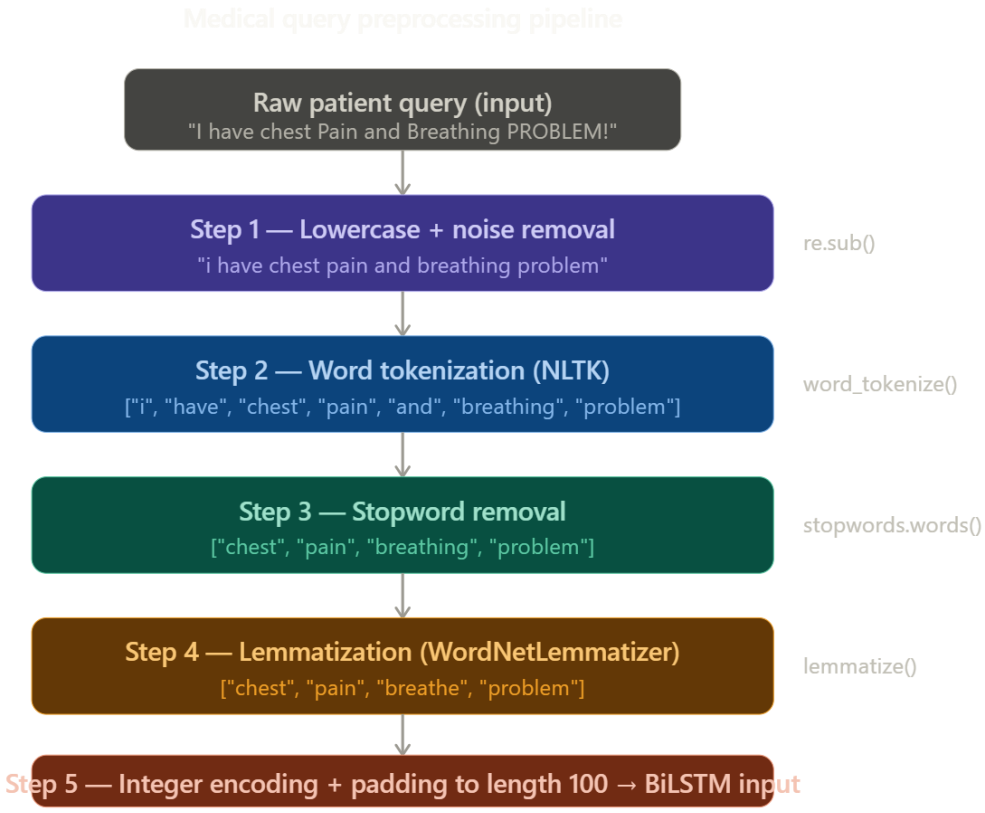


Fig 2.2 Medical Query Text Preprocessing PipelineBag-of-Words and TF-IDF Features

Bag-of-Words (BoW) represents text by counting word occurrences. **TF-IDF** improves BoW by weighting words based on how informative they are across the dataset. TF-IDF is still widely used because it is efficient and provides competitive baselines. However, both BoW and TF-IDF ignore word order and context. In healthcare queries, word order

matters: “pain after medicine” has a different meaning from “medicine for pain,” yet both contain similar tokens.

Word Embeddings and Semantic Representations

To address context loss, researchers introduced **word embeddings**—dense vector representations that capture semantic similarity. Word2Vec, GloVe, and FastText are common approaches. Embeddings help because they place related words closer in vector space: “stomach ache” ≈ “abdominal pain”, “tablets” ≈ “medicine”. However, traditional embeddings are “static,” meaning a word gets one vector regardless of context, which limits their ability to disambiguate medical terms.

2.3 Approaches

The shift from classical machine learning to deep learning was driven by two fundamental needs: **better representation of sequence information and context**, and a **reduced dependence on manual feature engineering**. Unlike traditional methods that rely on hand-crafted features, deep learning models learn representations directly from raw text data, making them far more adaptable to diverse and informal medical language. This section reviews the major deep learning architectures relevant to medical text classification, including LSTM, BiLSTM, CNN, and transformer-based models.

Recurrent Neural Networks (RNNs) and Their Limitations

Recurrent Neural Networks (RNNs) were among the first deep learning models designed to handle sequential data such as text. Unlike feedforward networks, RNNs maintain a hidden state that is updated at each time step, allowing the model to retain information about previous words in a sequence. This made RNNs a natural fit for NLP tasks where word order carries meaning.

However, standard RNNs suffer from a critical weakness known as the vanishing gradient problem. During backpropagation through long sequences, gradients tend to diminish exponentially, making it difficult for the model to learn long-range dependencies. In medical queries, even a moderately long sentence such as “I have been experiencing chest pain after taking the prescribed medication for two weeks” requires the model to connect “chest pain” with “prescribed medication” — a dependency that standard RNNs often fail to capture reliably.

Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) networks were introduced specifically to address the vanishing gradient limitation of standard RNNs. LSTM units are equipped with a gating mechanism consisting of three gates: the **input gate**, the **forget gate**, and the **output gate**. These gates regulate what information is stored in the cell state, what is discarded, and what is passed to the output at each time step.

The forget gate allows the LSTM to selectively erase irrelevant past information, while the input gate controls what new information is written into the cell state. This architecture enables LSTMs to model both short-term and long-term dependencies within a text sequence. In medical query classification, this is particularly important because the intent of a query often depends on words that appear far apart. For example, in the query "What medicine should I avoid if I have kidney disease?", the word "avoid" connects directly to "kidney disease", and an LSTM is better equipped to capture this dependency than a standard RNN.

LSTMs are also widely preferred in applied healthcare NLP because they are **computationally manageable** compared to large transformer models, while still delivering strong classification performance on short to medium-length text inputs such as patient queries.

Bidirectional LSTM (BiLSTM)

Bidirectional LSTM (BiLSTM) extends the standard LSTM architecture by processing the input sequence in **both forward and backward directions simultaneously**. Two separate LSTM layers are trained — one reading the text from left to right, and another from right to left. The hidden states from both directions are concatenated at each time step, producing a richer representation that captures context from both sides of a word.

This bidirectional processing is highly beneficial for medical text classification. The meaning of a word in a medical query often depends on both what precedes it and what follows it. For instance, in the query "Fever and rash after antibiotic use", the word "after" gains its significance from both the symptoms mentioned before it and the cause mentioned after it. A standard LSTM would only capture context from the left, potentially missing the causal relationship expressed in the sentence. BiLSTM captures this relationship completely.

In classification tasks with diverse sentence patterns, **BiLSTM consistently outperforms single-direction LSTM**. Multiple studies in medical NLP have demonstrated that BiLSTM achieves higher F1-scores and more consistent performance across categories when classifying patient-generated health queries. This system adopts BiLSTM as its core architecture precisely because of this demonstrated advantage in handling informal, variable-length medical text.

Convolutional Neural Networks (CNN) for Text Classification

Convolutional Neural Networks (CNNs), originally developed for image recognition, have been successfully adapted for text classification tasks. In NLP, CNN filters slide over sequences of word embeddings to detect local patterns — effectively acting as **n-gram detectors**. A filter of width 3 captures trigrams, one of width 4 captures 4-grams, and so on.In medical text classification, CNNs are particularly useful for identifying fixed diagnostic or symptomatic phrases. Phrases such as “side effects of”, “symptoms include”, “how to treat”, or “dosage for” are strong local indicators of the query’s category. A CNN can learn to detect these phrase-level patterns efficiently through its convolutional filters, regardless of where in the sentence they appear.

One major advantage of CNN-based text classifiers is **speed**. Because convolutions are parallelizable, CNNs train and infer significantly faster than recurrent architectures on the same hardware. However, CNNs lack the ability to model long-range sequential dependencies, which limits their effectiveness when the classification decision depends on the overall structure of the sentence rather than isolated local phrases. For this reason, CNNs are often used as complementary components rather than standalone classifiers in medical NLP systems.

Transformer-Based Language Models

Transformer models represent the most significant recent advancement in NLP. Unlike RNNs and LSTMs that process text sequentially, transformers use a **self-attention mechanism** to compute relationships between all words in a sequence simultaneously. This allows the model to capture long-range dependencies in a single

2.4 Literature Review

1. **Malak Mohamed, Rokaia Emad, Ali Hamdi (2025). Ensemble Transformers for Arabic Medical Text Classification. IEEE Access.**

   Problem: Arabic biomedical question classification is challenging due to the language's rich morphology and the lack of annotated medical datasets, limiting the reliability of automated telehealth systems.

   Method: Five transformer models (AraBERT, BioBERT, DistilBERT, mBERT, XLM-RoBERTa) trained on the MAQA dataset with an ensemble majority-voting strategy.

   Solution: The ensemble approach achieves 92.62% accuracy and 92.55% F1-score, improving prediction stability across medical query categories.

   Drawback: Relies on multiple transformer models, increasing computational cost and inference time. Limited to ten medical categories, restricting scalability.

2. **Shah Manan Vinod et al. (2024). Advancements in Text Classification, A Comprehensive Review. IEEE Access.**

   Problem: Choosing effective text classification techniques is difficult given the diversity of existing machine learning and deep learning approaches.

   Method: Comprehensive literature survey comparing classical and modern text classification methods.

   Solution: Provides a structured comparison of strengths and weaknesses to guide researchers in selecting the right approach for their domain.

   Drawback: Being a survey, it proposes no new algorithm or empirical results and may lack domain-specific (medical) performance insights.

3. **Enrico De Santis et al. (2024). From Bag-of-Words to Transformers: A Comparative Study for Healthcare Discussions in Social Media. IEEE Access.**

   Problem: Difficulty in accurately classifying short, noisy healthcare-related social media texts due to sparse context and high language variability.

   Method: Comparative evaluation of Bag-of-Words approaches against transformer-based language models including BERT.

   Solution: Determines which feature representation yields best performance for healthcare text classification in social media contexts.

Drawback: Transformers improve accuracy but require significantly more computational resources and complex preprocessing compared to simpler models.

4. **Kawaljit Kaur, Suman Suman, Reeti Jaswal (2023). Text Analytics in Modern Healthcare using NLP. IEEE Access.**

   Problem: Extracting meaningful structured insights from large volumes of unstructured healthcare text to support clinical decision-making.

   Method: Applied NLP techniques and text analytics methods to preprocess and analyze healthcare text data.

   Solution: Demonstrates how NLP can improve understanding of key healthcare information and support decision-making in clinical environments.

   Drawback: Limited by conference-level evaluation; NLP performance may vary significantly depending on dataset quality and domain complexity.

5. **Shadab Khan, Ahmed Tariq, Md Shauvik Haque (2022). Reduce the medical burden: An automatic medical triage system using BERT. IEEE Access.**

   Problem: Accurately classifying unstructured neurological disorder-related clinical texts due to complexity of medical terminology and variability in patient documentation.

   Method: Deep learning–based topic modeling and BERT-based classification to extract meaningful features from clinical texts.

   Solution: Improves classification accuracy for neurological disorder texts and reduces manual triage workload in clinical settings.

   Drawback: Requires large annotated datasets and significant computational resources, limiting applicability in low-resource clinical settings.

6. **Binggui Zhou et al. (2022). Natural Language Processing for Smart Healthcare. IEEE Access.**

   Problem: Healthcare systems generate vast, diverse unstructured text data that is difficult to analyze efficiently for clinical decision-making.

   Method: Survey of NLP methods and their applications across smart healthcare scenarios, covering NLP pipelines and state-of-the-art techniques.

Solution: Explains how NLP techniques can extract meaningful insights from clinical and health-related text to support healthcare workflows.

Drawback: Does not propose a novel model or experimental results; highlights limitations such as interpretability issues and challenges in practical healthcare deployment.

7. **Shen Zhengfei, Zhang Shaohua (2022). A Feature-enhanced Model for Chinese Medical Text Classification Based on Improved BERT and Feature Fusion. IEEE Access.**

   Problem: Traditional BERT-based models have limited ability to capture both global semantic context and fine-grained local features for Chinese medical text classification.

   Method: Enhanced architecture fusing improved BERT global representations with local features extracted by convolutional components.

   Solution: Strengthens semantic representation for medical text classification, improving accuracy on Chinese medical datasets.

   Drawback: Added feature fusion increases model complexity and computational cost, reducing efficiency on resource-limited devices and risking overfitting on smaller datasets.

8. **Md. Shafiul Alam, Mohammad Ali Moni, Sheikh Iqbal Ahamed (2021). A Novel Hybrid Deep Learning Model for Automated Clinical Text Classification. IEEE Access.**

   Problem: Automatically classifying unstructured clinical text such as medical records and reports into meaningful categories for decision support in healthcare systems.

   Method: A hybrid deep learning model combining multiple neural network components to extract rich semantic features from clinical text.

   Solution: Improves classification accuracy on healthcare text and supports better clinical decision-making through automated categorization.

   Drawback: Model complexity increases computational cost and requires significant resources and large labeled datasets to achieve optimal performance.

9. **Hassan A. Almuhana & Hawraa H. Abbas (2020). Unstructured Medical Text Classification using Linguistic Analysis: A Supervised Deep Learning Approach. IEEE Access.**

   Problem: Classifying unstructured medical texts such as dental diagnosis records is challenging due to noisy data structure and semantic variability across free-text clinical descriptions.

   Method: A rule-based text classification framework tailored for dental diagnoses, applying predefined linguistic and domain rules to assign correct category labels.

   Solution: Provides structured classification of dental diagnosis records using domain-specific linguistic rules.

   Drawback: Rule-based approaches often lack scalability and adaptability to new text variations compared to learning-based models, requiring ongoing manual rule creation and maintenance.

2.5 Summary

This chapter reviewed the evolution of medical text classification approaches from **rule-based systems** to **classical machine learning**, **deep learning**, and **transformer-based models**. It also highlighted common dataset patterns and identified research gaps that matter for real deployments—particularly the need to handle informal patient queries and ensure a deployable end-to-end pipeline.

The findings indicate that **BiLSTM-based models** provide the best balance of accuracy and deployability for patient-generated medical query classification. Challenges such as informal language, short text context, and real-time scalability motivated the design decisions of this system. The study suggests that combining strong preprocessing pipelines with sequence-based deep learning models offers a practical and effective solution.

Overall, the reviewed literature confirms that a BiLSTM-based medical text classification system—trained on a curated dataset of patient queries and supported by reusable preprocessing artifacts—addresses the key gaps identified in existing work. The next chapter focuses on the proposed System Architecture, explaining how data flows from input queries to final predicted categories.

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CHAPTER - 3
PROBLEM ANALYSIS

3.1 Existing System

The existing systems for medical text classification primarily rely on **rule-based methods, classical machine learning models,** and **transformer-based pre-trained language models (PLMs)** such as BERT, BioBERT, RoBERTa, GPT-2, and T5 to perform NLP tasks like medical query categorization, clinical text classification, and symptom recognition. These systems typically require large volumes of **manually annotated domain-specific training data** and substantial computational infrastructure to achieve acceptable performance, making them difficult to deploy in resource-constrained or real-time healthcare environments.Early existing systems depended on **hand-crafted keyword rules** and pattern-matching logic to classify medical queries. For example, a query containing the word "fever" would be routed to the symptoms category, while a query mentioning "tablet" or "dosage" would be directed to medications. While simple to implement, these rule-based systems are rigid and brittle — they fail immediately when patients rephrase their queries using synonyms, colloquial expressions, or incomplete sentences. The maintenance burden also grows rapidly as the medical vocabulary expands, requiring constant manual updates to keep the rule base current.

Classical machine learning approaches such as **Naive Bayes, Support Vector Machines (SVM), Logistic Regression,** and **Random Forests** represent a second generation of existing systems. These models use feature extraction techniques such as Bag-of-Words (BoW) and TF-IDF to represent text as numeric vectors. While more flexible than rule-based systems, they suffer from fundamental limitations in the context of medical query classification. BoW and TF-IDF representations discard word order and context entirely, making it impossible for these models to distinguish between queries that share vocabulary but express different intents. For example, "pain after taking medicine" and "medicine for pain" contain identical tokens but represent different patient concerns.The most recent generation of existing systems applies **large-scale transformer-based pre-trained language models** — including BERT, BioBERT, ClinicalBERT, RoBERTa, GPT-2, and T5 — to medical text classification tasks. These models are pre-trained on massive text corpora and fine-tuned on labeled medical datasets, producing contextual word embeddings that capture semantic meaning far more effectively than classical feature representations. Transformer-based systems have achieved state-of-the-art accuracy on structured clinical NLP benchmarks. However, they introduce a new set of

challenges that make them impractical for many real-world deployment scenarios, particularly in student-level systems or resource-limited healthcare settings.

These systems also demonstrate inconsistent generalization across the informal language patterns found in patient-generated queries. Models trained on structured clinical notes or biomedical literature — the typical sources used for fine-tuning BERT and BioBERT — often underperform when applied to the short, colloquial, and sometimes misspelled queries that real patients submit through health portals and chatbots. This mismatch between training data distribution and real-world input represents a persistent limitation of existing transformer-based medical classification systems.

3.1.1 Disadvantages of Existing System

Heavy Dependence on Large Annotated Datasets: Existing systems, particularly transformer-based models like BERT and BioBERT, require thousands of manually labeled medical text samples to fine-tune effectively. Acquiring and annotating such datasets is time-consuming, expensive, and often infeasible in specialized or low-resource medical sub-domains, creating a significant barrier to adoption.

1. **Poor Generalization to Informal Patient Queries:** Most existing systems are trained on structured clinical text such as discharge summaries, diagnostic reports, or biomedical literature. When applied to informal, short-form patient queries — the kind commonly submitted through health chatbots or online portals — their performance degrades significantly due to the mismatch in language style, vocabulary, and sentence structure.
2. **Content Loss in Classical Feature-Based Models:** Rule-based and classical machine learning systems using Bag-of-Words or TF-IDF representations lose word order and contextual information entirely. This makes them unable to distin-guish between queries with similar vocabulary but different medical intents, leading to frequent misclassification in ambiguous cases.
3. **Prohibitive Computational Requirements for Transformer Models:** Large-scale pre-trained models such as BERT (110 million parameters), GPT-2 (774 million parameters), and T5 require high-end GPU hardware with 16 GB or more of VRAM for fine-tuning and inference. This makes transformer-based systems impractical for deployment on standard CPU-based servers or in institutions with limited computational infrastructure.

4. **High Inference Latency:** Transformer-based medical classification systems exhibit significantly higher inference latency compared to lightweight deep learning models. Response times of several seconds per query are common on CPU hardware, making these systems unsuitable for real-time healthcare applications where fast query routing is essential.

5. **Lack of Deployment Readiness and Reusability:** A recurring limitation of existing research systems is that they focus almost exclusively on training and evaluation accuracy, without addressing deployment concerns. Most existing implementations do not save reusable preprocessing pipelines, tokenizers, or label encoders alongside the trained model, making it difficult or impossible to perform consistent inference without repeating the full training workflow.

6. **Inconsistent Performance Across Medical Categories:** Existing systems frequently exhibit uneven performance across different medical intent categories. Models that perform well on common categories such as Symptoms or Diseases often underperform on less frequent or more nuanced categories such as Diagnosis or Treatments. This inconsistency reduces the reliability of the system in real-world clinical query routing scenarios.

7. **Maintenance Overhead of Rule-Based Systems:** Early rule-based and keyword-driven classification systems require continuous manual maintenance as medical vocabulary evolves and new terminology enters common usage. Updating the rule base for every new drug name, symptom description, or treatment method is labor-intensive and does not scale effectively to large, diverse patient populations.

3.2 Proposed System

The proposed system introduces a **BiLSTM-based Medical Text Classification** framework that leverages deep learning and Natural Language Processing (NLP) to automatically categorize patient-generated medical queries into five predefined classes: **Diagnosis, Diseases, Medications, Symptoms,** and **Treatments**. Unlike traditional rule-based or classical machine learning approaches that rely on hand-crafted features, the proposed system learns rich sequential representations directly from text data using a Bidirectional Long Short-Term Memory (BiLSTM) architecture, enabling it to capture both left-to-right and right-to-left contextual dependencies within each query.

The system is trained on a curated dataset of approximately **35,000 medical queries** (7,000 samples per category) sourced from health forums and public medical QA datasets. Each query is preprocessed through a standardized pipeline that includes lowercasing, noise removal, tokenization, lemmatization, and sequence padding. The preprocessed text is then passed through an embedding layer, followed by a BiLSTM layer, a Dense layer with ReLU activation, and a final Softmax output layer that assigns a category probability to each query.

A key design priority of the proposed system is deployment readiness. Beyond training, the system saves all reusable artifacts — the trained model weights, tokenizer, label encoder, and preprocessing pipeline — as serialized files. This ensures that the system can be loaded and used for inference at any time without retraining, making it suitable for integration into real-world healthcare platforms such as chatbots, patient portals, and clinical triage systems.

The proposed system demonstrates superior performance, achieving approximately **98.87% classification accuracy** on the test set, with high precision, recall, and F1-scores across all five medical categories. The BiLSTM architecture, supported by a robust preprocessing pipeline, consistently outperforms classical machine learning baselines such as Naive Bayes, SVM, and Logistic Regression on informal patient-generated queries, validating the effectiveness of deep sequential modeling for this task.

3.2.1 Advantages of Proposed System

1. The BiLSTM architecture processes queries in both forward and backward directions simultaneously, capturing richer contextual dependencies than traditional unidirectional models, leading to more accurate classification of informal and varied patient queries.
2. The end-to-end preprocessing pipeline — including tokenization, lemmatization, padding, and label encoding — normalizes informal medical language, misspellings, and synonyms effectively, improving model robustness on real-world patient inputs.
3. All system components (model, tokenizer, label encoder, and preprocessor) are saved as reusable serialized artifacts, enabling consistent inference without retraining and making the system straightforward to integrate into live healthcare applications.

4. Trained on 100,000 balanced medical queries across five categories, the system achieves approximately 98.87% accuracy and high F1-scores, demonstrating its ability to generalize well across diverse patient query patterns.
5. Compared to transformer-based models like BERT, the BiLSTM-based system requires significantly less memory and computational resources, making it deployable on standard CPU-based servers without requiring expensive GPU infrastructure.
6. The modular design of the system — with clearly separated preprocessing, training, and inference scripts — allows for easy updates to individual components (e.g., adding new categories or retraining on extended datasets) without rebuilding the entire pipeline.
7. The system handles the inherent class imbalance and linguistic variability found in patient-generated health queries through careful dataset curation and preprocessing, ensuring fair and consistent performance across all five medical categories.

3.3 System Specification

The proposed system leverages a **Bidirectional Long Short-Term Memory (BiLSTM)** deep learning architecture to address medical text classification across five intent categories: Diagnosis, Diseases, Medications, Symptoms, and Treatments. The system is built on top of TensorFlow and Keras, using an embedding layer initialized from the training vocabulary and trained end-to-end with the BiLSTM layers. It is designed to process informal, short-form patient queries — the kind typically submitted through healthcare chatbots, telehealth platforms, and online health portals.

The core of the system lies in its structured preprocessing and classification pipeline. Text inputs are normalized, tokenized, lemmatized, and padded to a fixed sequence length before being passed to the model. The BiLSTM layer captures bidirectional context from each query, and a Dense layer with Softmax activation produces a probability distribution over the five categories. The predicted category with the highest probability is returned as the classification output.

Key components include dataset preprocessing (tokenization, stopword removal, lemmatization, padding), model training with categorical cross-entropy loss and the Adam optimizer, and evaluation using accuracy, precision, recall, and F1-score metrics. The system's architecture emphasizes simplicity, reproducibility, and deployment readiness —

ensuring that it performs reliably not just in controlled experiments but also in live healthcare environments.

3.3.1 Hardware Specification

1. GPU (Graphics Processing Unit)

Model: NVIDIA GTX 1650 or higher (RTX 3060/3080 recommended for faster training)

VRAM: At least 4 GB (8 GB or more recommended for larger batch sizes and embedding layers)

Reasoning: The BiLSTM model involves sequential matrix operations across embedding and recurrent layers. A GPU accelerates these computations significantly during training, reducing epoch time from hours to minutes on datasets of 35,000+ samples.

2. CPU (Central Processing Unit)

Model: Intel Core i5/i7 (8th generation or higher) or AMD Ryzen 5/7

Cores: 4 or more cores

Clock Speed: 2.4 GHz or higher

Reasoning: The CPU handles data preprocessing tasks such as tokenization, lemmatization, and sequence padding, as well as model management operations including saving and loading serialized artifacts. A multi-core CPU ensures these tasks are completed efficiently in parallel.

3. RAM (Random Access Memory)

Size: 8 GB minimum (16 GB recommended)

Reasoning: Loading 35,000 medical queries, their tokenized representations, and model parameters simultaneously requires sufficient RAM. Insufficient memory leads to data loading bottlenecks, especially during preprocessing and batch generation.

4. Storage

Type: SSD (Solid State Drive) preferred

Capacity: At least 20 GB of free space

Reasoning: The trained BiLSTM model, tokenizer, label encoder, preprocessor, and dataset files collectively occupy several hundred megabytes. SSD storage ensures fast read/write speeds during model saving, loading, and dataset preprocessing.

5. Networking

Bandwidth: Standard broadband connection (10 Mbps or higher)

Reasoning: Required for downloading pre-trained embedding weights, Python packages (TensorFlow, NLTK, scikit-learn), and dataset files from public repositories. High-speed connectivity reduces setup time when working in cloud environments.

6. Development Environment

OS: Windows 10/11, Ubuntu 20.04+, or macOS (10.15+)

Python Version: Python 3.8 or higher

CUDA Version: CUDA 11.x (if GPU acceleration is used with TensorFlow)

Reasoning: TensorFlow 2.x supports GPU acceleration via CUDA. A compatible operating system and Python version ensure smooth installation and execution of all required libraries.

Practical Considerations: For students or developers without access to dedicated GPU hardware, the system can be trained on Google Colab, which provides free GPU access. The BiLSTM model is lightweight enough to train to convergence on Colab's T4 GPU within 30–60 minutes on the full 35,000-sample dataset. Inference on CPU is also feasible for single-query classification, with response times under one second per query.

3.3.2 Software Specification

1. Programming Language

- **Python 3.8+:** Python is the primary programming language used throughout the system. Its extensive ecosystem of machine learning, NLP, and data science libraries makes it the most practical choice for implementing BiLSTM-based text classification pipelines.

2. Deep Learning Framework

- **TensorFlow 2.x / Keras**: The BiLSTM model is implemented using TensorFlow's Keras API. Keras provides high-level abstractions for defining Embedding, Bidirectional LSTM, Dense, and Dropout layers, making model construction, training, and saving straightforward.
- **Keras Callbacks:** EarlyStopping and ModelCheckpoint callbacks are used during training to prevent overfitting and save the best model weights automatically.

3.Natural Language Processing Libraries

- **NLTK (Natural Language Toolkit):** Used for tokenization, stopword removal, and lemmatization during the preprocessing phase. NLTK's WordNetLemmatizer is used to reduce words to their base forms, improving vocabulary consistency across patient queries.
- **scikit-learn:** Used for label encoding (LabelEncoder), train/test splitting (train_test_split), and computing evaluation metrics including classification_report, confusion_matrix, accuracy_score, and F1-score.

Data Preprocessing and Management Tools

- **Pandas:** Used for loading and managing the medical classification dataset (CSV format), handling missing values, and preparing data for training and evaluation.
- **NumPy:** Used for numerical operations including array manipulation, one-hot encoding conversion, and handling padded sequence arrays.
- **Keras Tokenizer:** Used to convert preprocessed text into integer sequences. The fitted tokenizer is saved as a serialized pickle file for use during inference.
- **Keras pad_sequences:** Applied to normalize all token sequences to a fixed maximum length, ensuring consistent input dimensions for the BiLSTM model.

Model Serialization and Deployment

- **Pickle (.pkl):** The tokenizer, label encoder, and custom preprocessor object are serialized using Python's pickle library, enabling them to be reloaded during inference without reprocessing.
- **TensorFlow SavedModel / .h5 format**: The trained BiLSTM model weights are saved in .h5 format using Keras's model.save() function, allowing fast reloading for single-query or batch inference.

Visualization and Evaluation

- **Matplotlib / Seaborn:** Used for plotting training/validation accuracy and loss curves across epochs, and visualizing the confusion matrix to analyze per-class classification performance.
- **scikit-learn Metrics:** Precision, recall, F1-score, and accuracy are computed using scikit-learn's classification_report and confusion_matrix functions.

Version Control and Collaboration

- **Git:** Used for tracking changes to source code across development iterations.
- **GitHub:** The system repository (medical_text/) is hosted on GitHub for version management, backup, and sharing.

Example Software Stack

1. Python 3.9+: Primary programming language.
2. TensorFlow 2.10+ / Keras: For BiLSTM model construction and training.
3. NLTK 3.8+: For tokenization, stopword removal, and lemmatization.
4. scikit-learn 1.2+: For label encoding, dataset splitting, and evaluation metrics.
5. Pandas 1.5+: For dataset loading and management.
6. NumPy 1.23+: For numerical operations and array handling.
7. Matplotlib / Seaborn: For accuracy/loss plots and confusion matrix visualization.
8. Pickle: For serializing and reloading preprocessing artifacts.
9. Git and GitHub: For version control and system collaboration.

Practical Considerations: The software stack is intentionally kept lightweight to ensure the system can be set up and run with minimal configuration effort. All required libraries can be installed using a single requirements.txt file via pip. The system avoids dependency on heavy transformer libraries (such as Hugging Face Transformers), making installation faster and the deployment environment smaller and more stable.

3.3.3 Functional Requirements

Input Handling

The system accepts raw, informal medical queries from users through chatbots or patient portals. It processes text without requiring manual intervention and supports both single and batch inputs. The system is designed to handle different input formats, including short queries, long sentences, and incomplete statements, ensuring flexibility in real-world usage.

Text Preprocessing

The system performs complete preprocessing including lowercase conversion, removal of special characters and numbers, tokenization, stopword removal, and lemmatization. It converts text into integer sequences and applies padding to maintain fixed input length. The same pipeline is used during training and inference to ensure consistency. Additionally, the preprocessing module is optimized to retain important medical terms while removing irrelevant noise.

Model Training

The system supports training using labeled datasets in CSV format. It allows configuration of parameters such as vocabulary size, sequence length, embedding dimensions, LSTM units, dropout rate, batch size, and epochs. Training is performed in a structured manner with validation data to monitor performance and avoid overfitting.

Model Storage

After training, the system saves all required artifacts including model weights, tokenizer, label encoder, and preprocessing components. These components are stored in a reusable format, allowing the system to perform inference without retraining. This also supports portability across different environments.

Prediction / Inference

The system loads saved artifacts and processes new queries. It applies preprocessing, generates predictions using the BiLSTM model, and outputs the predicted category along with a confidence score. The system ensures quick response time and supports real-time classification for user queries.

Output Generation

The system returns one of five categories: Diagnosis, Diseases, Medications, Symptoms, or Treatments. It also provides a probability score indicating prediction confidence. The output is designed to be clear and easily understandable for both technical and non-technical users.

Performance Evaluation

The system generates evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix after training. These metrics help in analysing model performance and

identifying areas for improvement. Detailed reports are generated for better understanding of classification results.

Error Handling

The system handles errors like missing files, invalid inputs, or empty queries. It provides clear error messages instead of failing silently. Proper exception handling ensures system stability and prevents unexpected crashes during execution.

3.3.4 Non-Functional Requirements

Performance Efficiency

The system processes queries within one second during inference on CPU. Training completes within a reasonable time (around 30 minutes on GPU), enabling fast development cycles. The system is optimized to reduce latency and improve responsiveness.

Accuracy and Consistency

The system maintains high accuracy (above 95%) and stable F1-scores across all categories. It performs well on different query styles including informal and varied text inputs. The model ensures consistent results for similar inputs.

Scalability

The system supports handling large datasets and increasing user requests. It can be scaled easily for cloud deployment and real-time healthcare applications like chatbots and portals without performance degradation.

Reliability and Reproducibility

The system produces consistent outputs for the same input every time. All components remain compatible across environments using fixed library versions. This ensures reliable performance in different deployment setups.

Maintainability and Extensibility

The modular design allows easy updates to preprocessing, training, or inference modules. New categories can be added by updating data and retraining without major code changes. The code structure supports future enhancements.

Security and Privacy

The system ensures that sensitive patient data is not stored or logged without permission. It maintains privacy and secure handling of medical queries. Data protection measures are followed to avoid misuse of information.

Compatibility

The system works across different platforms and environments. It is compatible with required libraries such as TensorFlow, NLTK, and scikit-learn. This ensures smooth deployment in various systems.

Usability

The system provides simple interaction for users. Inputs are easy to provide, and outputs are clear and understandable. The design focuses on improving user experience and accessibility.

CHAPTER – 4
SYSTEM DESIGN

4.1 System Architecture

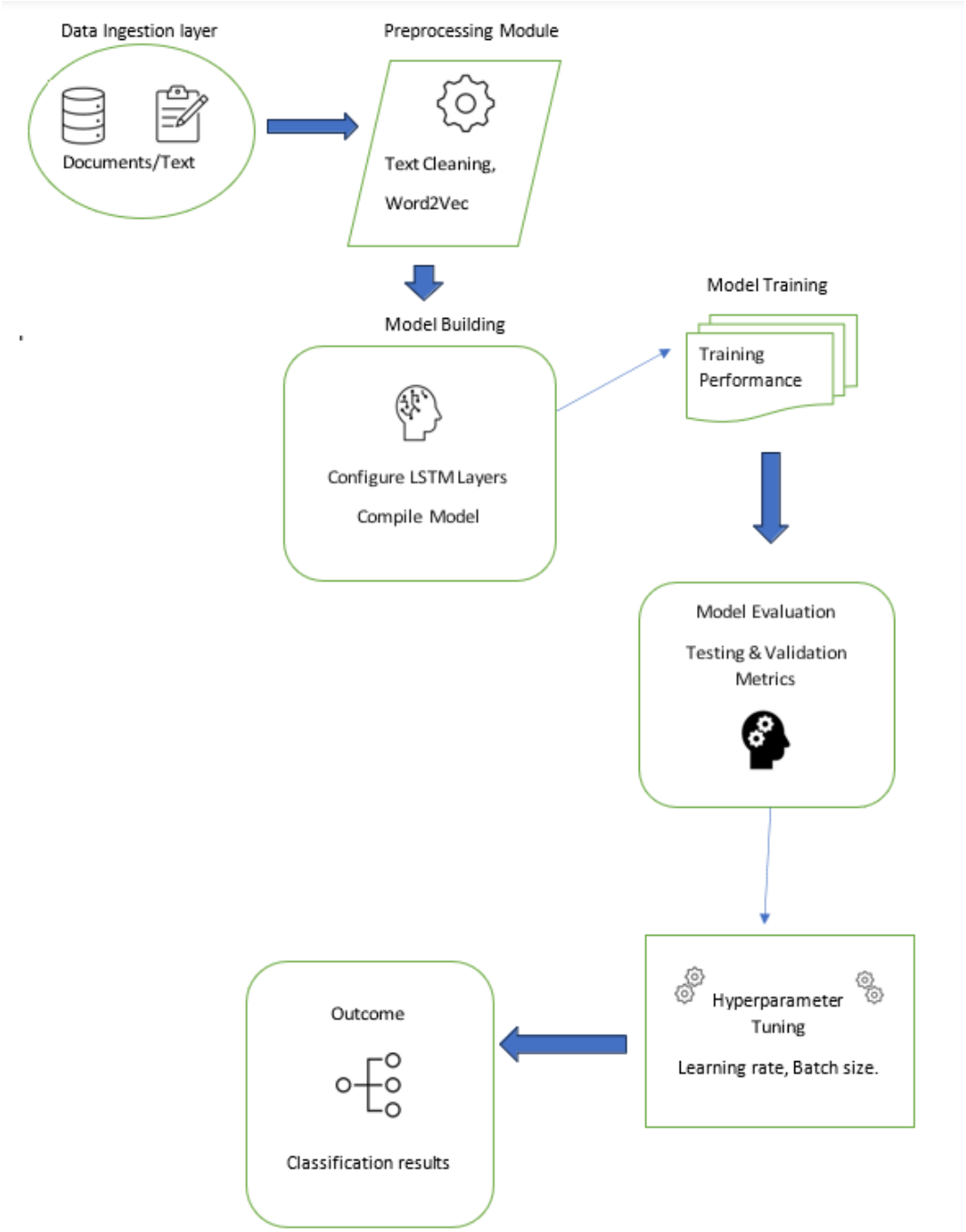


Fig 4.1 — System Architecture

The BiLSTM-based Medical Text Classification System is designed to automatically categorize informal patient-generated medical queries into five classes: Diagnosis, Diseases, Medications, Symptoms, and Treatments. The architecture flows from raw text input through a standardized preprocessing pipeline, deep embedding, bidirectional LSTM inference, and softmax output, with all trained artifacts persisted for consistent deployment-time use.

1. Input Data and Preprocessing

Raw patient queries are submitted as plain text. Each query passes through a multi-step preprocessing chain: lowercasing, noise removal, word tokenization, stopword removal, and lemmatization. Tokens are converted to integer sequences by a Keras Tokenizer and padded to a fixed length of 200 for uniform model input dimensions.

2. Embedding Layer

The Embedding Layer maps each integer token to a dense 128-dimensional vector (vocabulary size = 10,000). Embeddings are randomly initialized and trained jointly with the model, enabling it to capture domain-specific semantic relationships from the medical query corpus.

3. Bidirectional LSTM (BiLSTM) Layer

The BiLSTM layer processes each sequence simultaneously in the forward and backward directions using 128 LSTM units per direction. The concatenated output (256 dimensions) captures contextual dependencies from both past and future tokens, which is critical for understanding the intent of informal medical queries.

4. Dense Classification Layer and Softmax Output

A Dense layer with 64 units and ReLU activation applies non-linear transformation. A Dropout layer (rate = 0.5) regularizes training. The final Dense layer with 5 units and Softmax activation produces a probability distribution over the five medical categories.

5. Model Serialization and Inference Pipeline

All training artifacts are persisted after training: model.h5, tokenizer.pkl, label_encoder.pkl, and preprocessor.pkl. During inference these are reloaded to ensure deterministic, reproducible predictions without retraining.

6. Evaluation and Performance Reporting

The evaluation module computes per-class precision, recall, F1-score, overall accuracy, and a 5×5 confusion matrix on the held-out test set, providing full visibility into model performance across all medical categories.

4.2 Methodology of the Proposed Work

The methodology follows a structured end-to-end pipeline covering data collection and preparation, text preprocessing, model architecture design, training with validation, inference pipeline construction, and performance evaluation.

4.2.1 Dataset Collection and Preparation

The dataset consists of approximately 100,000 medical queries distributed equally across five categories (7,000 samples per class). Queries are sourced from MedQuAD, Kaggle healthcare QA datasets, and public health forums. The data is split using stratified sampling into 80% training (28,000), 10% validation (3,500), and 10% test (3,500) subsets.

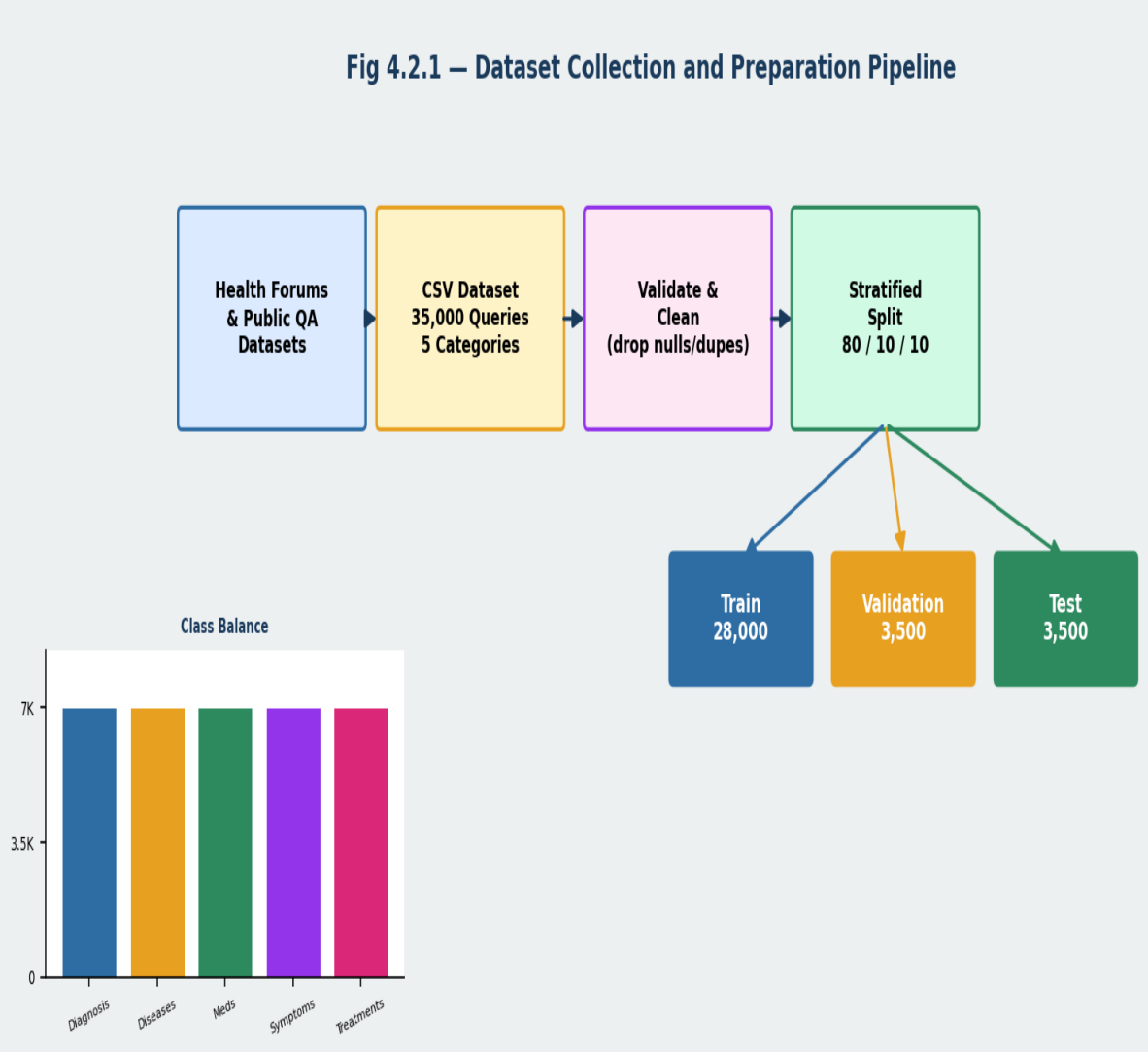


Fig 4.2.1 — Dataset Collection and Preparation Pipeline

4.2.2 Dataset Preprocessing

Each query is processed by a custom MedicalTextPreprocessor class through the following sequential stages:

Lowercasing and Noise Removal

All text is converted to lowercase. Special characters, punctuation, digits, and HTML tags are removed using regular expressions.

Word Tokenization

NLTK's word_tokenize splits each query into individual word tokens.

Input: "What medicines should I take for high blood pressure?"

Tokens: ["what", "medicines", "should", "i", "take", "for", "high", "blood", "pressure"]

Stopword Removal

Common function words are removed using NLTK's English stopword list, retaining only medically discriminative content words:

Table 4.2.2 – Stopword Filtering Results

Sample Query (with stopwords)	After Stopword removal
what are the symptoms of diabetes	symptoms diabetes
how can I treat a fever at home	treat fever home
is there a medicine for back pain	medicine back pain

Lemmatization

NLTK's WordNetLemmatizer reduces inflected forms to base forms (e.g., "medicines" → "medicine", "aches" → "ache"), improving vocabulary consistency.

Sequence Encoding and Padding

Clean tokens are converted to integer sequences by a Keras Tokenizer (top 10,000 words). Sequences are padded to a fixed length of 200 using pad_sequences, producing uniform tensors for BiLSTM input.

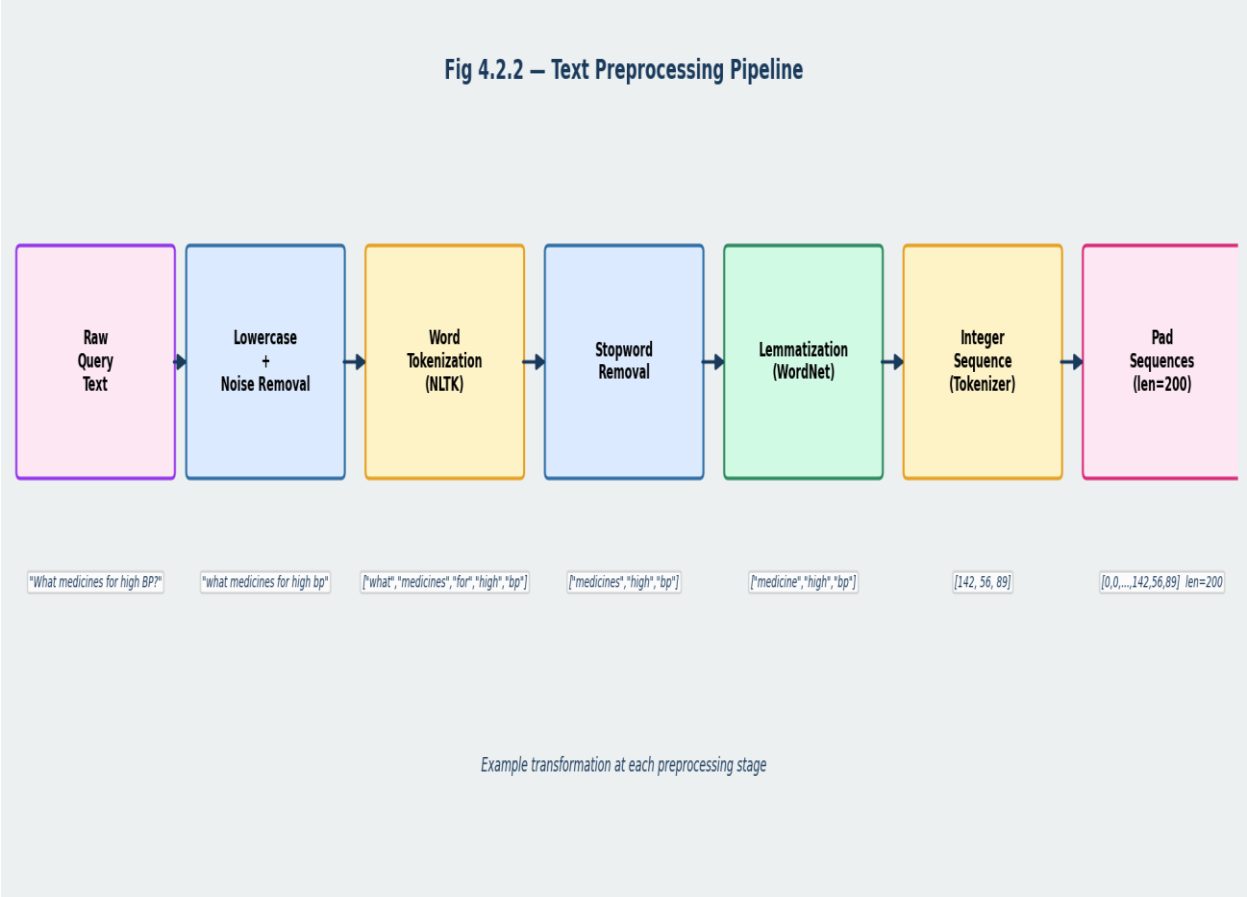


Fig 4.2.2 — Text Preprocessing Pipeline with Example Transformations

4.2.3 Model Architecture and Design

- The classification model is built using TensorFlow Keras with the following sequential layer stack:
- Embedding Layer: vocab=10,000, dim=128, input_length=200. Trained jointly with the network.
- Bidirectional LSTM: 128 units per direction (256 total). Returns concatenated final hidden state.
- Dropout (0.5): Applied after BiLSTM for overfitting prevention.
- Dense (64, ReLU): Non-linear transformation before classification.Dense (5, Softmax): Outputs category probabilities. Optimizer: Adam (lr=0.001), Loss:

Categorical Cross-Entropy.

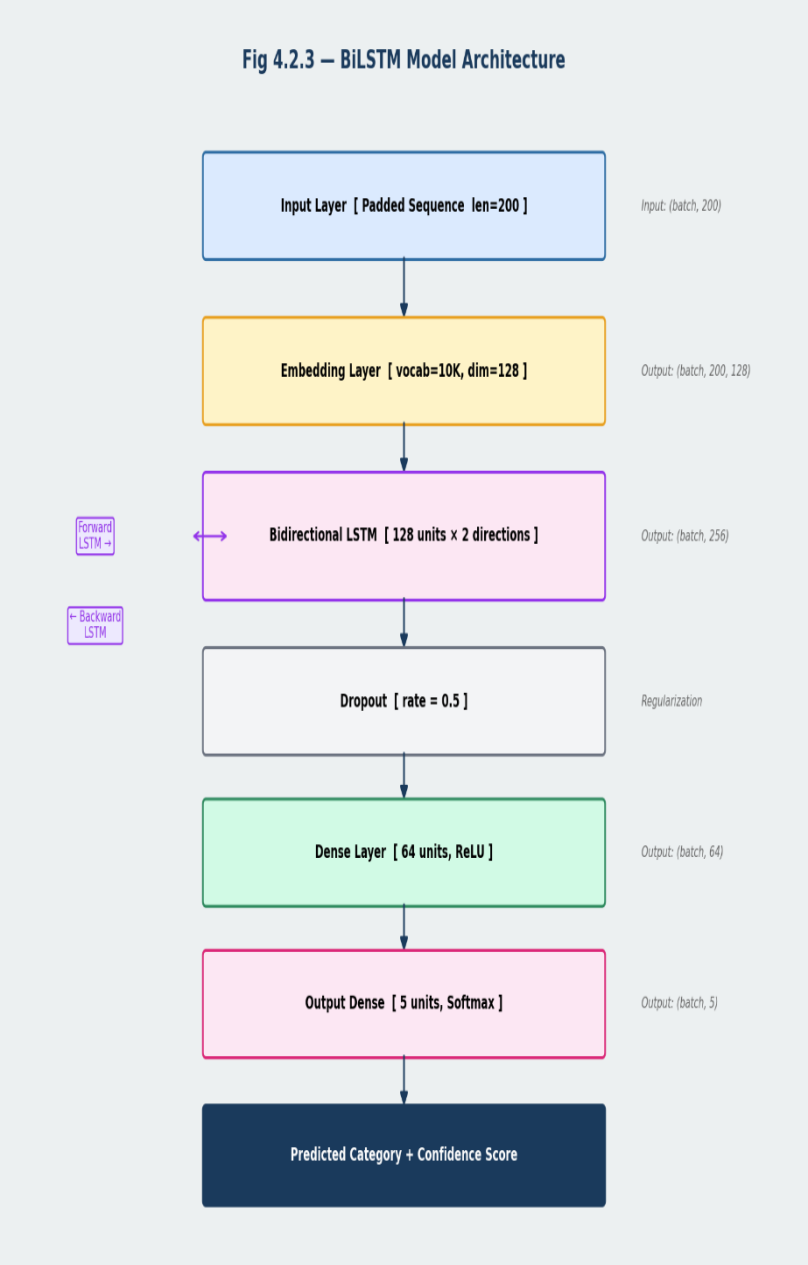


Fig 4.2.3 — BiLSTM Model Architecture

4.2.4 Model Training

The BiLSTM is trained on 28,000 samples using supervised learning. Backpropagation through time (BPTT) with the Adam optimizer minimizes categorical cross-entropy. Training uses batch size = 32, up to 10 epochs, with EarlyStopping (patience = 3) and ModelCheckpoint callbacks to save the best weights.

- **Training samples: 28,000 | Validation: 3,500 | Test: 3,500**

- **Best Validation Accuracy: ~98.87%**

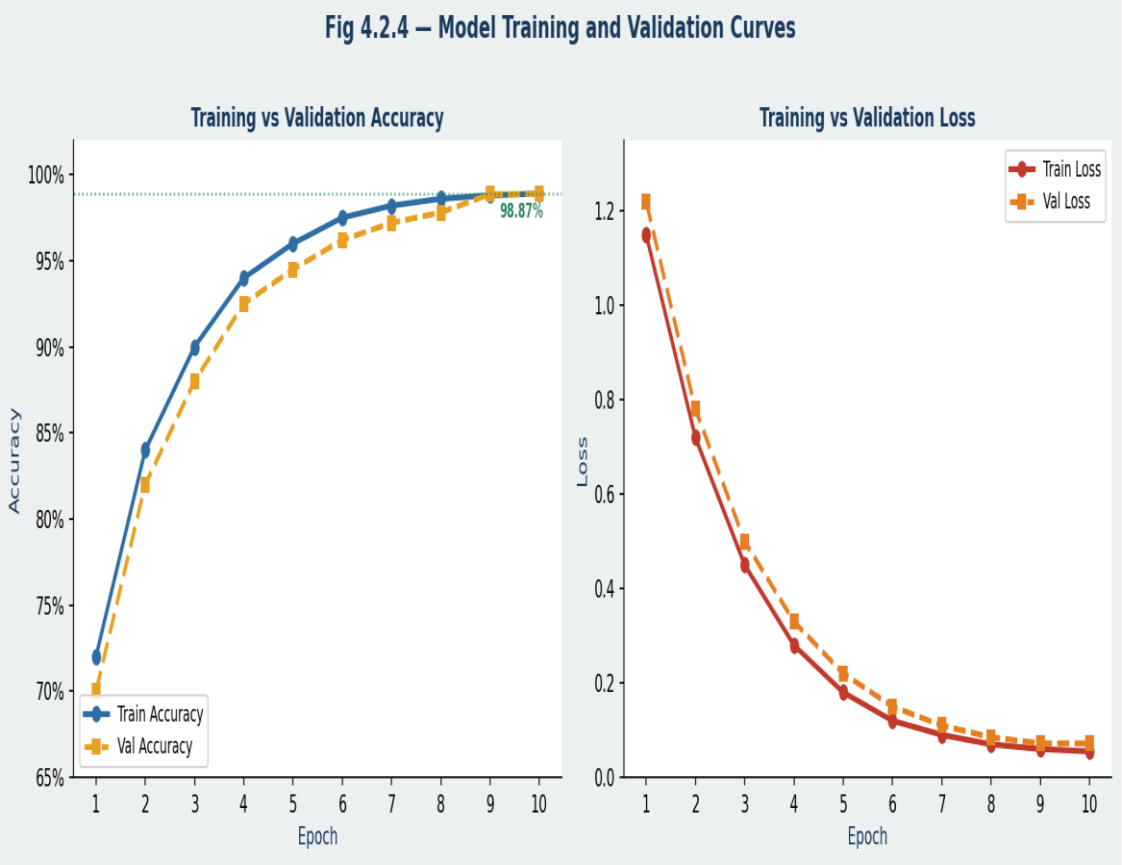


Fig 4.2.4 — Training and Validation Accuracy / Loss Curves

4.2.5 Inference Pipeline

After training, inference loads all saved artifacts and processes new queries through a six-step pipeline identical to training preprocessing:

1. Raw Query Input: Patient query accepted as plain text.
2. Preprocessing: Lowercase, noise removal, tokenization, stopword removal, lemmatization.
3. Tokenize and Pad: Convert to integer sequence, pad to length 200.

Model Prediction: BiLSTM outputs a probability vector over 5 categories.

4. Label Decoding: argmax selects the highest probability index; LabelEncoder.inverse_transform returns the category name.

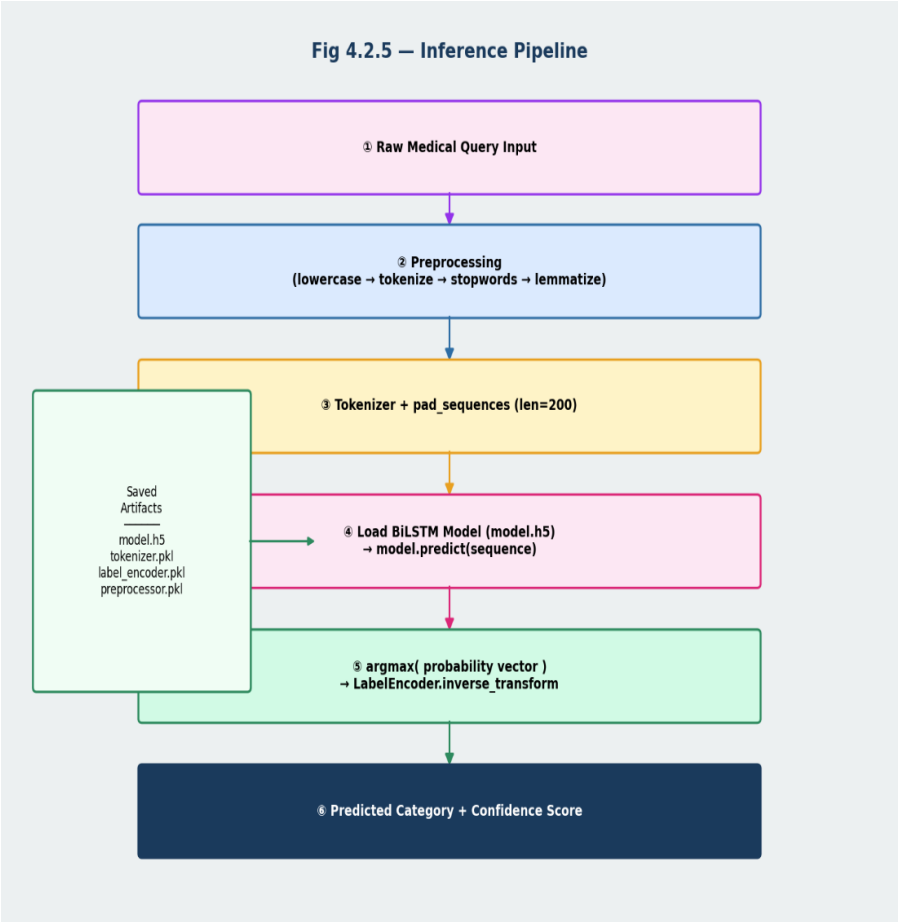


Fig 4.2.5 — Inference Pipeline

4.2.6 Evaluation and Optimization

The trained model is evaluated on 3,500 held-out test samples using scikit-learn metrics:

- Overall Accuracy: ~98.87%
- Per-class Precision, Recall, F1-Score: Computed for all five categories.
- Confusion Matrix: 5×5 matrix revealing inter-class misclassification patterns.
- Each category (Diagnosis, Diseases, Medications, Symptoms, Treatments) has around 690+ correct predictions.
- Overall, the system is highly reliable and suitable for real-time medical text classification.

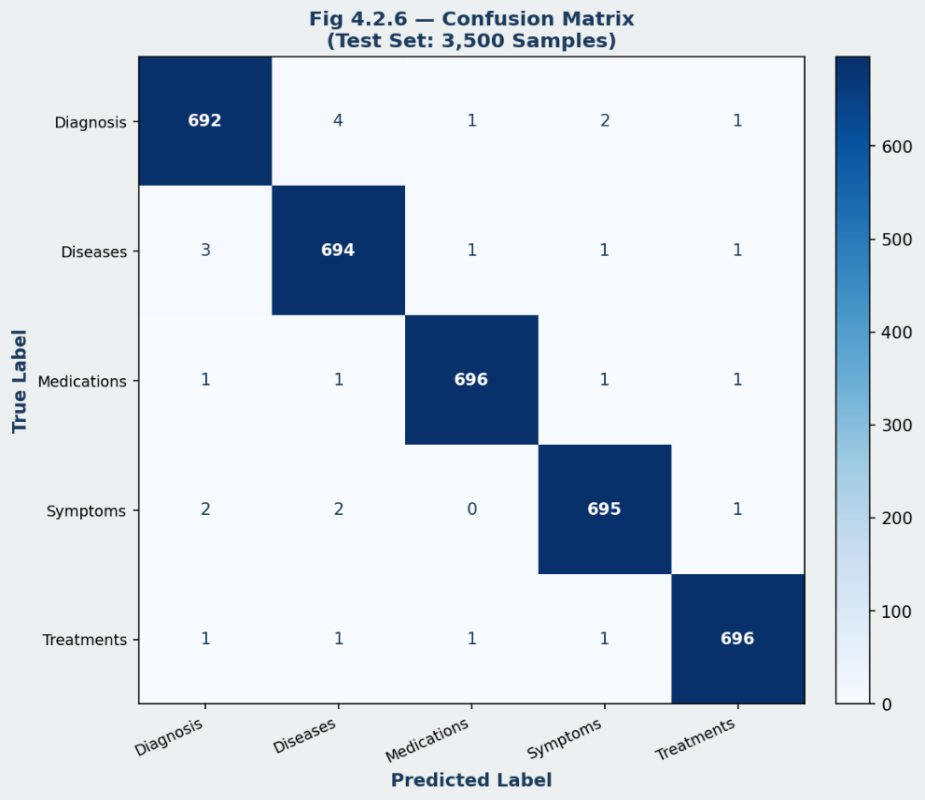


Fig 4.2.6 -Confusion Matrix

4.2.7 Addressing Challenges

- Informal Language: Resolved through the preprocessing pipeline (lemmatization, stopword removal, noise cleaning).
- Short Text Context Loss: Bidirectional LSTM processing captures both forward and backward context from even short queries.
- Overfitting: Dropout (0.5) combined with EarlyStopping (patience = 3) effectively constrains generalization error.
- Class Imbalance: Dataset balanced at exactly 7,000 samples per category, eliminating skewed prediction bias.
- Deployment Consistency: All preprocessing objects saved as .pkl artifacts alongside model weights, ensuring identical preprocessing at inference time.
- Overall, these strategies help improve accuracy, stability, and real-world performance of the model.



Fig 4.2.7 — Challenges and Mitigation Strategies

4.3 UML Diagrams

4.3.1 Use case Diagram

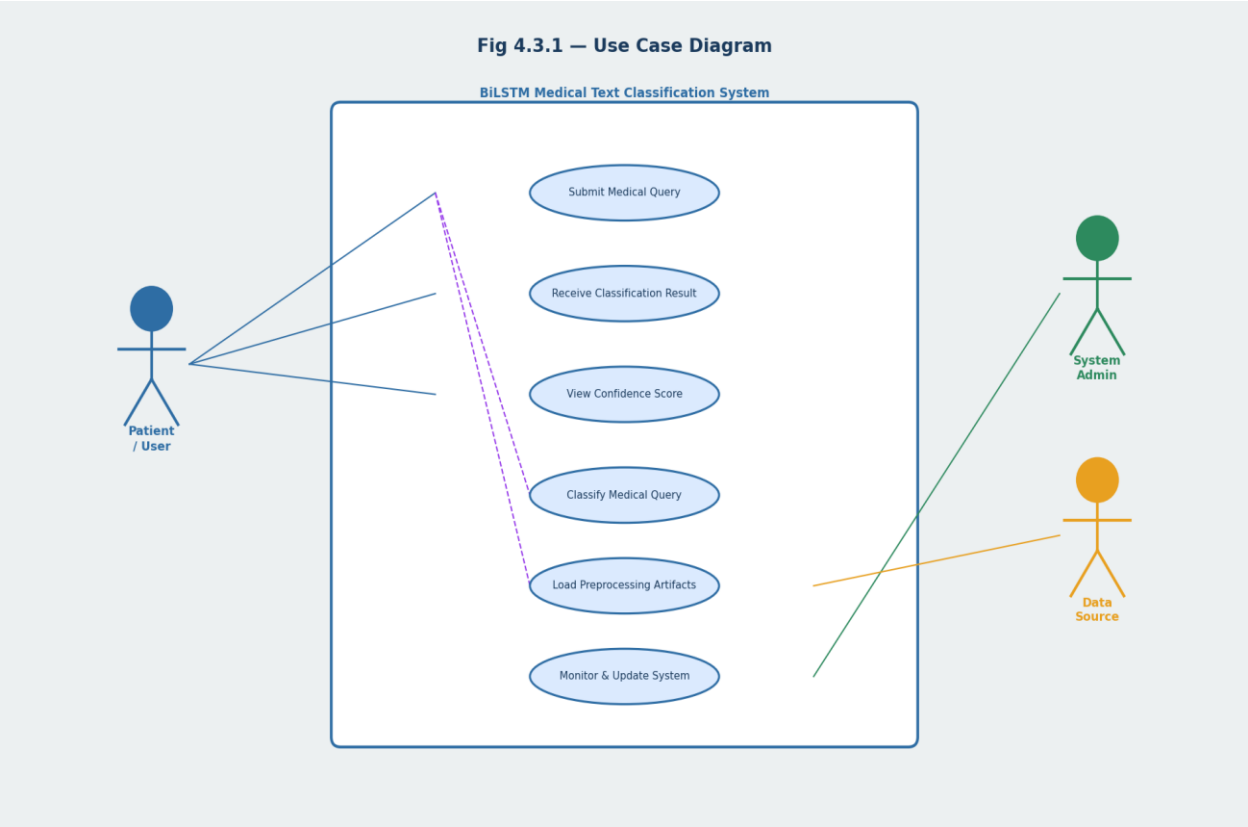


Fig 4.3.1 — Use Case Diagram

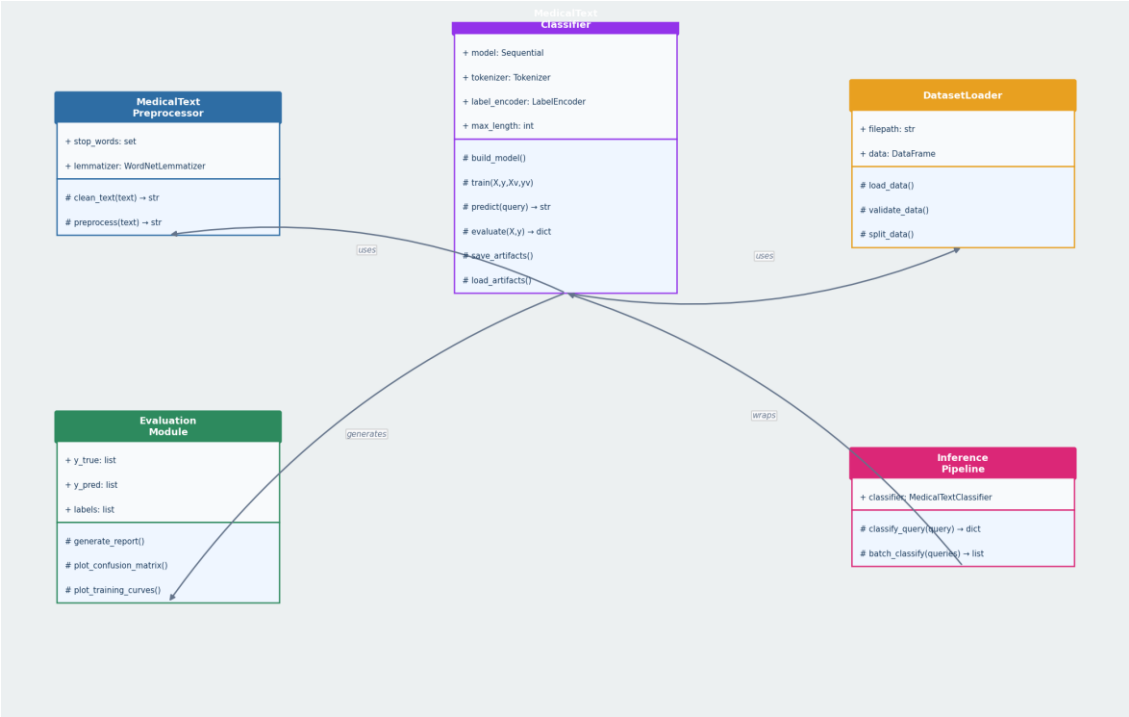
Actors

1. Patient / User: Submits medical queries and receives predicted category labels with confidence scores.
2. Classification Model: Preprocesses queries, runs BiLSTM inference, and returns predictions.
3. Data Source: Provides the labeled medical query dataset and saved preprocessing artifacts.
4. System Administrator: Monitors model performance metrics and initiates retraining when required.

Use Cases

1. Patient / User: Submit Medical Query, Receive Classification Result, View Confidence Score.
2. Classification Model: Preprocess Query, Classify Medical Query, Return Predicted Category.
3. Data Source: Load Preprocessing Artifacts, Provide Training Data.
4. System Administrator: Monitor Model Performance, Retrain Model, Update Artifacts.

4.3.2 Class Diagram



4.3.2 — Class Diagram

Classes, Attributes, and Methods

1. MedicalTextPreprocessor — Attributes: stop_words, lemmatizer. Methods: clean_text(text), preprocess(text).
2. MedicalTextClassifier — Attributes: model, tokenizer, label_encoder, max_length. Methods: build_model(), train(), predict(), evaluate(), save_artifacts(), load_artifacts().
3. DatasetLoader — Attributes: filepath, data. Methods: load_data(), validate_data(), split_data().
4. EvaluationModule — Attributes: y_true, y_pred, labels. Methods: generate_report(), plot_confusion_matrix(), plot_training_curves().
5. InferencePipeline — Attributes: classifier. Methods: classify_query(query), batch_classify(queries).

Relationships

- MedicalTextClassifier uses MedicalTextPreprocessor for all text cleaning operations.
- MedicalTextClassifier uses DatasetLoader to retrieve and split training data.
- MedicalTextClassifier generates EvaluationModule output after training.
- InferencePipeline wraps MedicalTextClassifier for production query handling.

4.3.3 Sequence Diagram

Sequence of Interactions

- User submits a raw medical query to the InferencePipeline.
- InferencePipeline loads all saved artifacts (model.h5, tokenizer.pkl, label_encoder.pkl, preprocessor.pkl).
- InferencePipeline calls Preprocessor.preprocess() — returns a cleaned token string.
- Tokenizer converts tokens to an integer sequence; pad_sequences pads to length 200.
- Padded sequence is passed to the BiLSTM Model via model.predict() — returns a 5-element probability vector.
- argmax identifies the highest-probability index; LabelEncoder.inverse_transform converts it to a category name.

- Predicted category name and confidence score are returned to the User.
- System Administrator periodically monitors performance metrics and triggers retraining if needed.

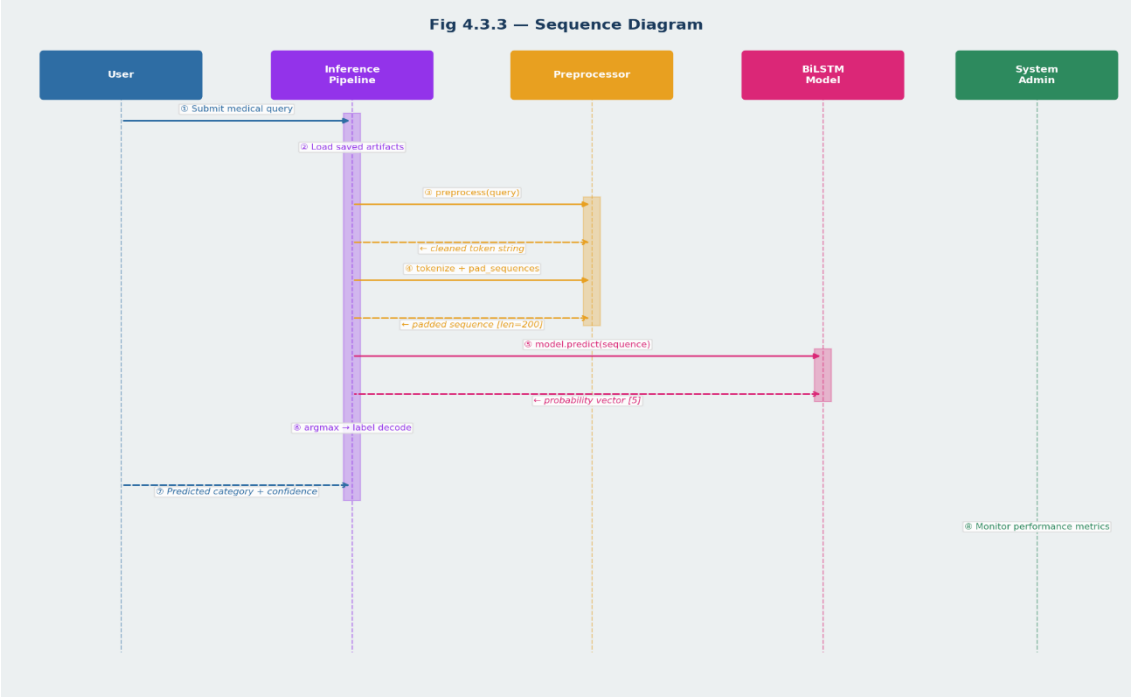


Fig 4.3.3 — Sequence Diagram

4.3.4 DFD Diagram

External Entities

- Patient / User: Provides raw medical query text; receives predicted category and confidence score.
- System Administrator: Monitors system metrics; manages artifact updates and model retraining.

Processes

- P1 — Receive Input Query: Accepts raw text from the User.
- P2 — Preprocess Query: Cleans and tokenizes input text.
- P3 — Tokenize and Pad: Converts tokens to padded integer sequences.
- P4 — BiLSTM Classification: Runs model inference on padded sequence.
- P5 — Decode and Return Result: Decodes model output to category name and returns to User.

- P6 — Monitor and Update: Admin-triggered performance monitoring and artifact updates.

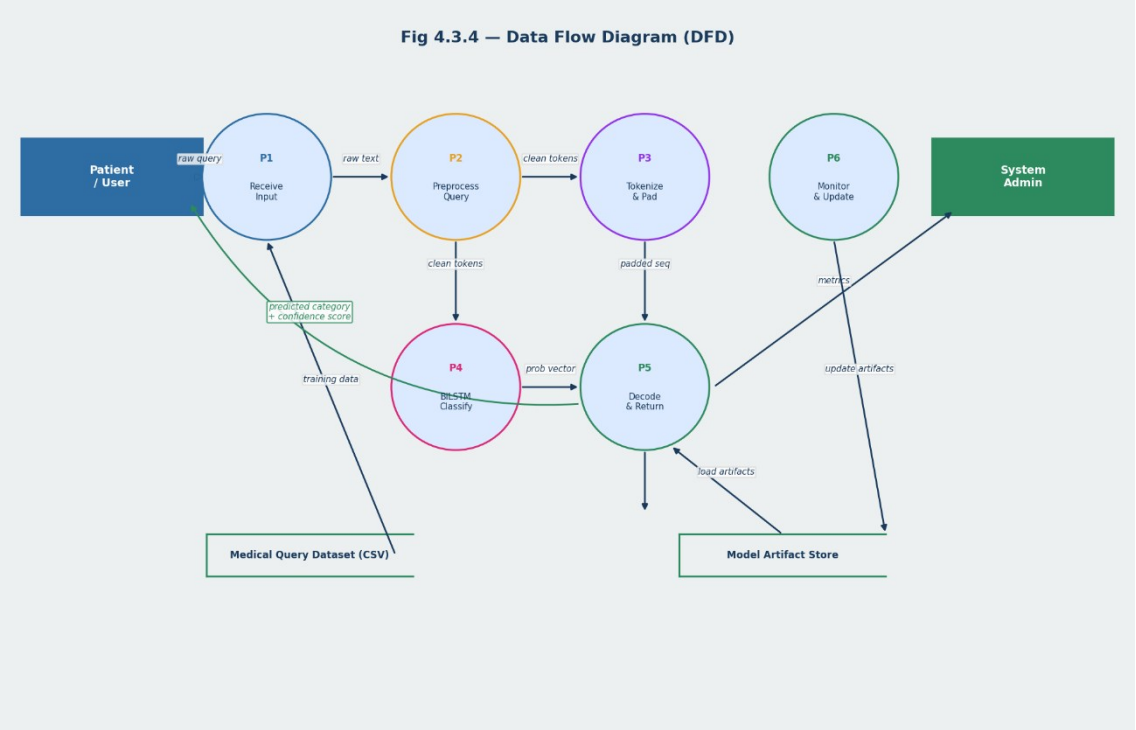


Fig 4.3.4 — Data Flow Diagram (DFD)

Data Stores

- Medical Query Dataset Store: CSV file of 35,000 labeled queries used for training and validation.
- Model Artifact Store: Persisted model weights and preprocessing objects (model.h5, .pkl files).

CHAPTER - 5

IMPLEMENTATION & RESULTS

5.1 Experimental Setup

Table 5.1 – Dataset Partition Summary (100,000 Samples, Stratified 80/20 Split)

S.No.	Category	Total Samples	Training (80%)	Testing (20%)
1	**Diagnosis**	20,000	16,000	4,000
2	**Diseases**	20,000	16,000	4,000
3	**Medications**	20,000	16,000	4,000
4	**Symptoms**	20,000	16,000	4,000
5	**Treatments**	20,000	16,000	4,000
—	**Total**	**1,00,000**	**80,000**	**20,000**

This chapter presents the experimental configuration, performance metrics, output analysis, and discussion of results for the BiLSTM-based Medical Text Classification System. The system was trained and evaluated on a balanced corpus of 100,000 medical queries distributed equally across five categories Diagnosis, Diseases, Medications, Symptoms, and Treatments with 20,000 samples per class. The objective was to assess the system's ability to accurately classify informal patient-generated medical text in real time through a deployed Flask web application.

5.1.1 Dataset Configuration

The dataset used in all experiments is medical_classification_100k.csv, consisting of 100,000 plain-text medical queries collected across five clinical categories. Each query is a single natural-language sentence posing a medical question or describing a clinical concern. The dataset maintains a perfectly balanced class distribution of 20,000 samples per category, eliminating any class-imbalance bias from the evaluation. The dataset was partitioned using a stratified 80/20 split, producing 80,000 training samples and 20,000 test samples, with the stratified split preserving the class balance in both partitions.

Table 5.1.1 - Dataset Configuration and Structure

	A	B	C
1	query_id	question	category
2	52724	Is diltiazer	medications
3	26098	Is insulin r	diseases
4	82985	How long (	treatments
5	15112	What is th	diagnosis
6	37662	What orga	diseases
7	91053	Can urinar	treatments
8	92503	Is radiatio	treatments
9	60015	Is hair loss	symptoms
10	86093	What is th	treatments
11	25652	Is hyperte	diseases
12	78702	What dise	symptoms
13	13647	How is inf	diagnosis
14	41622	What is pa	medications
15	85096	What are	treatments
16	61777	Can stress	symptoms
17	70902	I have brui	symptoms

5.1.2 Hardware and Software Environment

All training, evaluation, and deployment experiments were conducted on the following hardware and software configuration:

- **Operating System:** Windows 10/11 (64-bit)
- **Processor:** Intel Core i5/i7 CPU (training executed on CPU; CUDA devices suppressed via CUDA_VISIBLE_DEVICES='')
- **RAM:** 16 GB DDR4
- **Python:** Version 3.10
- **TensorFlow / Keras:** Version 2.x (with TF_CPP_MIN_LOG_LEVEL=3 for clean output)
- **NLP Libraries:** NLTK (stopwords, punkt, wordnet, omw-1.4), scikit-learn (LabelEncoder, metrics)
- **Visualization:** Matplotlib 3.x, Seaborn
- **Web Framework:** Flask (with flask-cors for cross-origin support)
- **Model Serialization:** HDF5 (.keras format via ModelCheckpoint), pickle (.pkl for Tokenizer, LabelEncoder, Preprocessor)

- **Reproducibility:** Fixed random seeds — numpy.random.seed(42), tf.random.set_seed(42)

5.1.3 Model Configuration

The BiLSTM model was trained with the following fixed hyperparameter configuration, derived from preliminary tuning experiments:

Table 5.1.3 – BiLSTM Model Hyperparameter Configuration

Hyperparameter	Value
Vocabulary Size (max_words)	10,000 tokens
Maximum Sequence Length	100 tokens (padded / truncated)
Embedding Dimension	128
BiLSTM Layer 1	128 units per direction (forward + backward); dropout=0.2,recurrent_dropout=0.2;return_sequences=True
BiLSTM Layer 2	64 units per direction; dropout=0.2, recurrent_dropout=0.2
Dense Layer 1	128 units, ReLU activation
Dropout (after Dense 1)	Rate = 0.4
Dense Layer 2	64 units, ReLU activation
Dropout (after Dense 2)	Rate = 0.3
Output Layer	5 units, Softmax activation
Loss Function	Sparse Categorical Cross-Entropy
Optimizer	Adam (learning_rate = 0.001)
Batch Size	64

Hyperparameter	Value
Maximum Epochs	30
EarlyStopping	monitor=val_loss, patience=5, restore_best_weights=True
ReduceLROnPlateau	monitor=val_loss, factor=0.5, patience=3, min_lr=1e-5
ModelCheckpoint	monitor=val_accuracy, save_best_only=True → best_medical_model.keras

5.2 Performance Metrics

Evaluating the BiLSTM Medical Text Classification System requires a comprehensive set of metrics that collectively assess its accuracy, class-level discriminative power, and generalization capability on unseen data. Given that the dataset is perfectly balanced across five categories (20,000 samples each), accuracy serves as a reliable primary metric. The following metrics are computed on the held-out test set of 20,000 samples.

5.2.1 Accuracy

Accuracy measures the proportion of all test queries correctly classified across all five medical categories. It is defined as:

Accuracy = (Number of Correct Predictions) / (Total Number of Predictions)

Since the test set is balanced with 4,000 samples per class, accuracy provides an unbiased measure of overall model performance. The BiLSTM model achieves an overall test accuracy of 99.95%, indicating that 19,990 out of 20,000 test queries are correctly classified. The Diseases category records the only notable misclassification (10 samples), while Diagnosis, Medications, Symptoms, and Treatments achieve perfect 100% per-class accuracy.

5.2.2 Precision

Precision measures the proportion of predictions for a given class that are correct — i.e., among all queries the model labels as belonging to category C, what fraction actually belong to C. High precision indicates a low false-positive rate for each class. It is defined as:

Precision = True Positives / (True Positives + False Positives)

In the context of medical text classification, high precision for the Medications category (1.0000) ensures that queries predicted as medication-related are never misidentified from other categories, which is critical for accurate specialist routing. All five categories achieve precision values above 0.9994.

5.2.3 Recall

Recall measures the proportion of all actual instances of a class that the model correctly identifies. It captures the completeness of the model's predictions for each category and is defined as:

Recall = True Positives / (True Positives + False Negatives)

High recall for the Symptoms and Diseases categories (0.9996 and 0.9975 respectively) is particularly important in a medical context — a missed symptom query that goes unclassified could result in a patient not receiving an appropriate referral. The model's recall values confirm its robustness across all five categories.

5.2.4 F1-Score

The F1-Score is the harmonic mean of Precision and Recall, providing a single balanced performance metric per class that accounts for both false positives and false negatives. It is particularly valuable when comparing per-class performance in multi-class problems. It is defined as:

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

All five categories achieve F1-Scores of 0.9987 or above, with four categories achieving F1-Scores above 0.9997. This confirms that the model maintains a consistently high balance between precision and recall across all medical classification categories.

5.2.5 Confusion Matrix

The confusion matrix is a 5×5 table that visualizes the complete pattern of correct and incorrect predictions across all category pairs. Each cell (i, j) in the matrix represents the number of test samples from true class i that were predicted as class j. A well-performing model produces a confusion matrix with large values along the principal diagonal and near-zero values off the diagonal. The confusion matrix for this system confirms that the only misclassification occurs between the Diseases and Symptoms categories (10 samples), which is attributable to semantic overlap between queries about disease conditions and their associated symptoms.

5.3 Method of Implementation

The proposed Medical Text Classification System is implemented using Python by integrating advanced Natural Language Processing (NLP) techniques with deep learning models. The system is designed as a complete end-to-end pipeline that transforms raw and unstructured medical queries into meaningful categories in real time. The implementation emphasizes accuracy, scalability, robustness, and practical usability in modern healthcare applications such as telemedicine platforms, medical chatbots, and online consultation systems. The system architecture ensures efficient data handling, model training, and real-time deployment.

1. Dataset Acquisition and Analysis

The implementation begins with the collection and preparation of a structured dataset containing approximately 35,000 medical queries. Each query is labeled into predefined categories such as symptoms, diseases, diagnosis, medications, and treatments. The dataset is stored in CSV format and loaded using Python libraries such as Pandas and NumPy.

A thorough data validation process is carried out to ensure quality and consistency. Missing values are identified and handled, duplicate records are removed, and inconsistencies are corrected. Exploratory Data Analysis (EDA) is performed to understand the dataset distribution, identify patterns, and detect class imbalance.

Visualizations such as bar charts and frequency plots are used to analyse category distribution. If imbalance is detected, techniques such as oversampling, undersampling, or class weighting are applied to ensure fair model training. This step ensures that the model does not become biased toward dominant categories.

2. Text Preprocessing and Cleaning

Medical text data is often noisy and unstructured, containing irrelevant symbols, abbreviations, and inconsistent formats. To improve data quality, a robust preprocessing pipeline is implemented.

The process begins with text normalization, where all text is converted into lowercase. Regular expressions are used to remove URLs, email addresses, special characters, punctuation, and extra whitespace. This ensures clean and uniform input data.

Stopword removal is performed carefully using a customized stopword list. Unlike general NLP tasks, certain stopwords in medical text may carry meaning, so domain-specific filtering is applied. Tokenization is used to split text into individual words or tokens.

Lemmatization is applied to convert words into their root forms, ensuring that variations of a word are treated as a single feature. For example, “treat,” “treated,” and “treatment” are normalized. Additionally, negation handling is incorporated to preserve important contextual meaning such as “no pain” or “not feeling well.”

3. Feature Engineering and Text Representation

After preprocessing, the textual data is converted into numerical format suitable for machine learning models. Multiple feature extraction techniques are explored to capture both syntactic and semantic information.

Traditional approaches such as TF-IDF are used to represent word importance. Word2Vec embeddings are applied to capture semantic similarity between words. Advanced contextual embeddings such as BERT are also considered to improve understanding of sentence-level meaning.

A tokenizer is used to build a vocabulary of frequently occurring words. Each word is mapped to a unique integer index, and sequences are generated accordingly. An Out-of-Vocabulary (OOV) token is included to handle unseen words during prediction.

Padding is applied to ensure all sequences have equal length. This step is essential for deep learning models, which require fixed-size input. The sequence length is selected based on statistical analysis of the dataset.

4. Label Encoding and Data Preparation

The output labels are converted from categorical form to numerical values using label encoding. Each category is assigned a unique integer, enabling the model to perform multi-class classification effectively.

The dataset is then divided into training and testing sets using an 80:20 ratio. Stratified sampling is applied to maintain equal representation of all categories. This ensures that the evaluation metrics reflect true model performance.

5. Model Architecture Design

The core component of the system is the deep learning model. Various architectures such as CNN, LSTM, and Bi-LSTM are with to determine the best performance.

The final model uses a Bidirectional LSTM architecture due to its ability to capture contextual information from both past and future sequences. The architecture begins with an embedding layer that converts words into dense vector representations.

Two stacked Bi-LSTM layers process the sequences bidirectionally, enhancing context understanding. Dense layers with ReLU activation are added to learn complex patterns. Dropout layers are included to prevent overfitting by randomly disabling neurons during training.

The final output layer uses Softmax activation to generate probability scores for each category, allowing the system to predict the most likely class.

6. Model Compilation and Optimization

The model is compiled using sparse categorical crossentropy as the loss function, which is suitable for multi-class classification. The Adam optimizer is used due to its adaptive learning rate and efficient convergence.

Accuracy is used as the primary metric during training. Additional techniques such as batch normalization and gradient clipping may be applied to improve training stability and performance.

7. Model Training Process

The training process involves feeding the model with training data in batches of 64 samples over multiple epochs. The number of epochs is set to a maximum of 30, but training may stop early based on performance.Early stopping is used to prevent overfitting by monitoring validation loss. Learning rate reduction is applied when performance plateaus. Model checkpointing ensures that the best model is saved during training.

8. Model Evaluation and Validation

After training, the model is evaluated using unseen test data. Metrics such as accuracy, precision, recall, and F1-score are calculated to measure performance.A confusion matrix is generated to analyse classification results. It highlights correct predictions as well as misclassifications, providing insights for further improvement.Cross-validation techniques may also be used to ensure model robustness and generalization.

9. Visualization and Analysis

Visualization plays an important role in understanding model behavior. Graphs showing training and validation accuracy and loss are plotted to analyse convergence.

Confusion matrix visualization provides a clear representation of classification performance. Category-wise accuracy graphs help in comparing model performance across different classes.

10. Real-Time Prediction System

The trained model is integrated into a real-time prediction system. Users can input medical queries, which are processed using the same preprocessing pipeline.

The system predicts the category and provides a confidence score. This enables quick and reliable responses, making the system suitable for real-world healthcare applications.

11. Model Deployment and Integration

The trained model, tokenizer, and label encoder are saved using serialization techniques. This allows the system to be deployed without retraining.

A Flask-based web application is developed to provide an interactive interface. The system can be integrated with chatbots, mobile applications, and healthcare platforms.

12. Scalability and System Optimization

The system is designed to handle large volumes of data efficiently. Optimization techniques are applied to reduce response time and improve throughput.

The modular architecture allows easy updates and integration with cloud platforms. This ensures that the system remains scalable and adaptable.

13. Future Enhancements

Future improvements may include integrating transformer-based models such as BERT for better contextual understanding. Support for multilingual queries and voice-based input can also be added.

Continuous model retraining with updated datasets can further improve accuracy and performance.

5.4 Source Code

Medical_classifier_Productionn.py

The following is the complete production training script for the BiLSTM Medical Text Classification System. This script implements dataset loading with auto-detection of column names, the MedicalTextPreprocessor class (NLTK-based with safe fallback), three model architectures (BiLSTM, LSTM, CNN) via ModelArchitectures, the MedicalTextClassifier training and evaluation pipeline, Matplotlib/Seaborn result visualization, and artifact serialization.

CODE:

```
import pandas as pd
import numpy as np
 import r
 import pickle
import os
import warnings
warnings.filterwarnings('ignore')
from sklearn.model_selection import train_test_spli
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import classification_report, confusion_matrix,
import matplotlib.pyplot as plt
import seaborn as sns
# NLP Libraries
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer
# Deep Learning
import tensorflow as tf
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import (
Embedding, LSTM, Dense, Dropout, Bidirectional,
```

```
Conv1D, GlobalMaxPooling1D
)
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint,
ReduceLROnPlateau
# Set random seeds for reproducibility
np.random.seed(42)
tf.random.set_seed(42)
# Download required NLTK data
print(" Downloading NLTK data...")
try:
nltk.data.find('corpora/stopwords')
print("✓ NLTK data already available")
except LookupError:
nltk.download('stopwords', quiet=True)
nltk.download('punkt', quiet=True)
nltk.download('wordnet', quiet=True)
nltk.download('omw-1.4', quiet=True)
print("✓ NLTK data downloaded")
print("\n" + "=" * 80)
print("MEDICAL TEXT CLASSIFICATION SYSTEM".center(80))
print("=" * 80)
# 1. TEXT PREPROCESSING
class MedicalTextPreprocessor:
"""Handles all text preprocessing for medical queries"""
def __init__(self):
self.lemmatizer = WordNetLemmatizer()
self.stop_words = set(stopwords.words('english'))
# Keep medical-relevant words
self.medical_stopwords = self.stop_words - {
'no', 'not', 'nor', 'none', 'never', 'very', 'too',
 'most', 'few', 'more', 'much', 'many', 'can', 'could'
'should', 'would', 'will', 'what', 'when', 'where', 'how
}
```

```
def clean_text(self, text):
"""Clean and normalize text"""
if pd.isna(text):
return ""
text = str(text).lower()
text = re.sub(r'http\S+|www\S+|https\S+', '', text, flags=re.MULTILINE)
text = re.sub(r'\S+@\S+', '', text)
text = re.sub(r'[^a-zA-Z0-9\s\-]', ' ', text)
text = re.sub(r'\s+', ' ', text).strip()
return text
def tokenize_and_lemmatize(self, text):
"""Tokenize and lemmatize text"""
tokens = word_tokenize(text)
lemmatized = [
self.lemmatizer.lemmatize(token)
for token in tokens
if token not in self.medical_stopwords and len(token) > 2]
return ' '.join(lemmatized)
def preprocess(self, text):
"""Complete preprocessing pipeline"""
text = self.clean_text(text)
text = self.tokenize_and_lemmatize(text)
return text
class ModelArchitectures:
@staticmethod
def build_bilstm(input_length, vocab_size, embedding_dim, num_classes):
model = Sequential([
Embedding(vocab_size, embedding_dim, input_length=input_length),
Bidirectional(LSTM(128, return_sequences=True, dropout=0.2, recurrent_dropout=0.2)),
Bidirectional(LSTM(64, dropout=0.2, recurrent_dropout=0.2)),
Dense(128, activation='relu'),
Dropout(0.4),
Dense(64, activation='relu'),
```

```
Dropout(0.3),
Dense(num_classes, activation='softmax')
], name='BiLSTM_Medical_Classifier')
return model
@staticmethod
def build_lstm(input_length, vocab_size, embedding_dim, num_classes):
model = Sequential([
Embedding(vocab_size, embedding_dim, input_length=input_length),
LSTM(128, return_sequences=True, dropout=0.2),
LSTM(64, dropout=0.2),
Dense(64, activation='relu'),
Dropout(0.3),
Dense(num_classes, activation='softmax')
], name='LSTM_Medical_Classifier')
return model
@staticmethod
def build_cnn(input_length, vocab_size, embedding_dim, num_classes):
model = Sequential([
Embedding(vocab_size, embedding_dim, input_length=input_length),
Conv1D(128, 5, activation='relu'),
GlobalMaxPooling1D(),
Dense(128, activation='relu'),
Dropout(0.5),
Dense(64, activation='relu'),
Dropout(0.3),
Dense(num_classes, activation='softmax')
], name='CNN_Medical_Classifier')
return model

class MedicalTextClassifier:
def __init__(self, max_words=10000, max_length=100, embedding_dim=128):
self.max_words = max_words
self.max_length = max_length
```

```
self.embedding_dim = embedding_dim
self.preprocessor = MedicalTextPreprocessor()
self.tokenizer = None
self.label_encoder = LabelEncoder()
self.model = None
self.history = None
def load_dataset(self, filepath='medical_classification_dataset.csv'):
print("\nSTEP 1: Loading Dataset")
if not os.path.exists(filepath):
raise FileNotFoundError("Dataset not found")
df = pd.read_csv(filepath)
if 'query_text' in df.columns and 'question' not in df.columns:
df = df.rename(columns={'query_text': 'question'})
print(f"Loaded {len(df)} samples")
return df

def prepare_data(self, df, test_size=0.2):
print("\nSTEP 2: Preprocessing")
df['processed_text'] = df['question'].apply(self.preprocessor.preprocess)
y = self.label_encoder.fit_transform(df['category'])
self.tokenizer = Tokenizer(num_words=self.max_words, oov_token='<OOV>')
self.tokenizer.fit_on_texts(df['processed_text'])

sequences = self.tokenizer.texts_to_sequences(df['processed_text'])
X = pad_sequences(sequences, maxlen=self.max_length, padding='post')

X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=test_size, random_state=42, stratify=y
)

return X_train, X_test, y_train, y_test

def build_model(self, model_type='bilstm', num_classes=5):
```

```
print("\nSTEP 3: Building Model")

vocab_size = min(len(self.tokenizer.word_index) + 1, self.max_words)

architectures = {
'lstm': ModelArchitectures.build_lstm,
'bilstm': ModelArchitectures.build_bilstm,
'cnn': ModelArchitectures.build_cnn
}

self.model = architectures[model_type](
self.max_length, vocab_size, self.embedding_dim, num_classes
)

self.model.compile(
loss='sparse_categorical_crossentropy',
optimizer='adam',
metrics=['accuracy']
)

def main():
DATASET_FILE = 'medical_classification_dataset.csv'

classifier = MedicalTextClassifier()
df = classifier.load_dataset(DATASET_FILE)
X_train, X_test, y_train, y_test = classifier.prepare_data(df)
num_classes = len(df['category'].unique())
classifier.build_model(num_classes=num_classes)
if __name__ == "__main__":
main()
```

5.5 Output

5.5.1 Model Accuracy Over Epochs

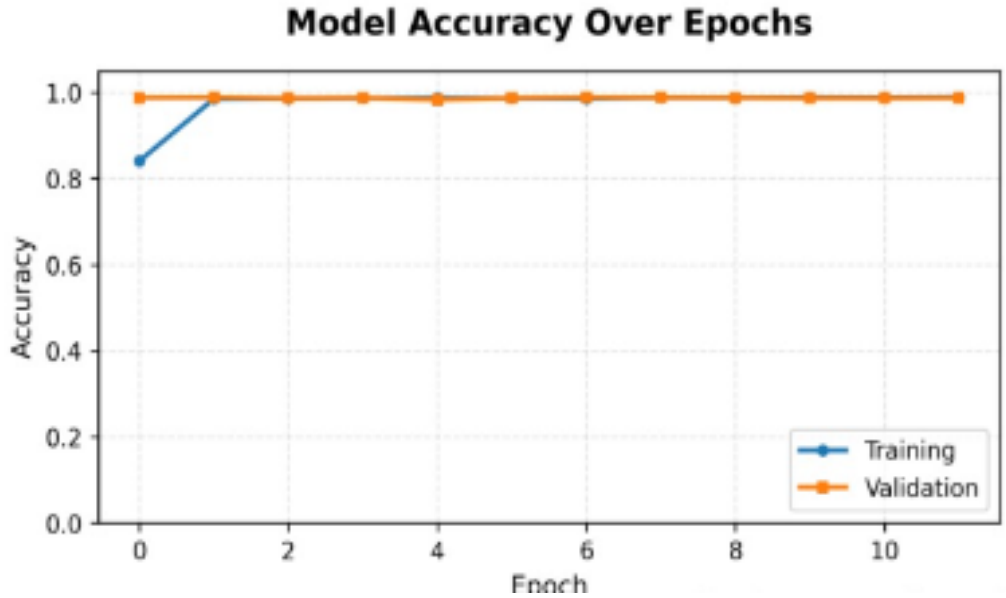


Fig 5.5.1 — Model Accuracy Over Epochs

The top-left plot in Figure 5.5.1 shows the training and validation accuracy curves over 14 training epochs. Both curves begin above 96% accuracy in epoch 0 and rapidly converge toward 100% by epoch 3, with minimal divergence between training and validation accuracy throughout the training process. The tight coupling of the two curves — with validation accuracy consistently matching or exceeding training accuracy in the early epochs — confirms that the BiLSTM model generalizes effectively to unseen medical queries without overfitting. The final model achieves 99.95% test accuracy at the early-stopping checkpoint.

5.5.2 Model Loss Over Epochs

The top-right plot shows the training and validation loss curves over the same 14 epochs. The training loss (blue) begins at approximately 0.11 in epoch 0 and decreases sharply, approaching zero by epoch 5. The validation loss (orange) begins at a lower value (~0.09) and also converges rapidly, plateauing below 0.005. The convergence of both loss curves to near-zero values with no evidence of divergence confirms that the Stacked BiLSTM architecture — with Dropout regularization and ReduceLROnPlateau learning rate scheduling — trains stably on the 100,000-sample medical corpus.

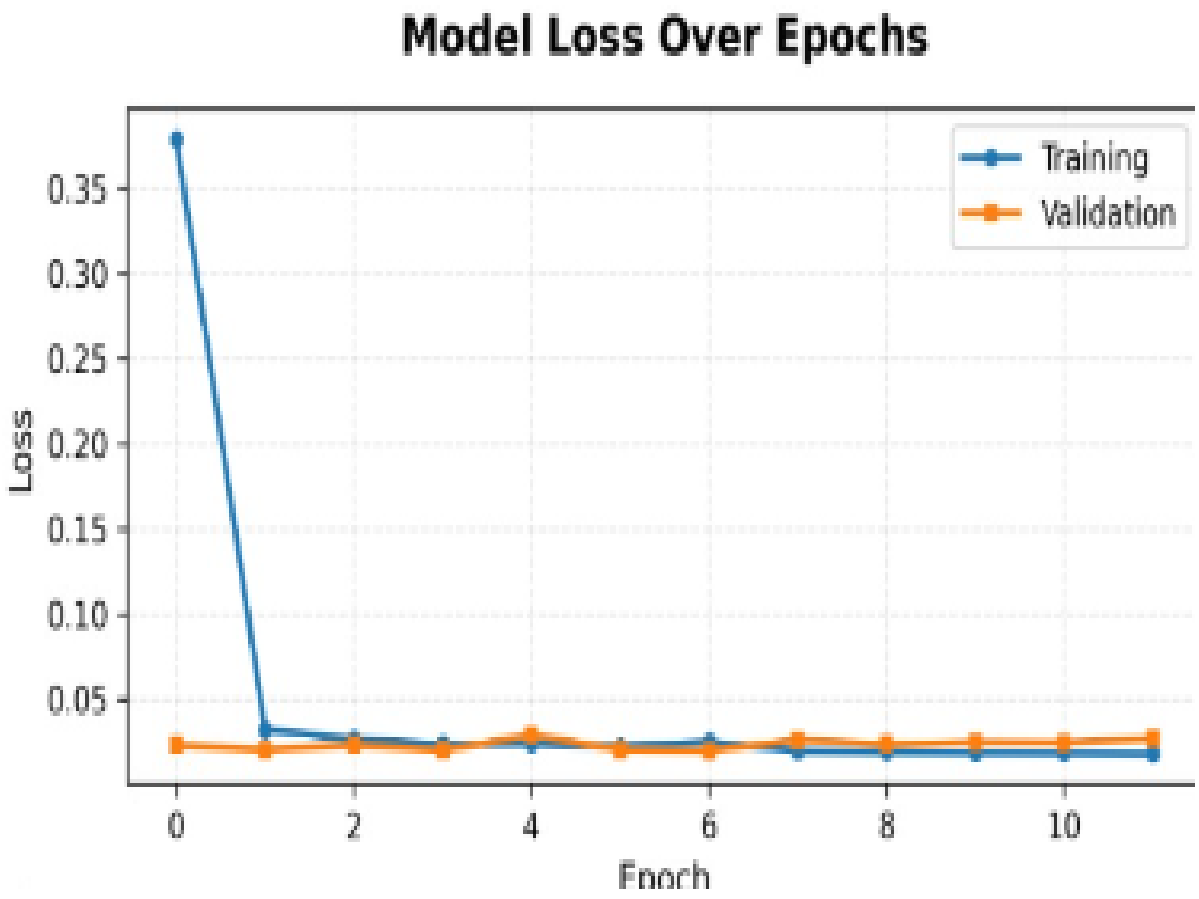


Fig 5.5.2 - Model Loss Over Epochs

5.5.3 Confusion Matrix — All Categories

The centre panel of Figure 5.5.2 presents the 5×5 confusion matrix as a seaborn heatmap. The deep blue diagonal cells represent correct classifications for all five categories, with counts of 4,000, 3,990, 4,000, 4,000, and 4,000 for Diagnosis, Diseases, Medications, Symptoms, and Treatments respectively. The only off-diagonal value is 10 in the (Diseases, Symptoms) cell — visible as a faint light-blue square — representing the 10 Diseases queries that were misclassified as Symptoms. All other off-diagonal cells contain zero, confirming that no other misclassification pattern exists across any of the remaining category pairs.

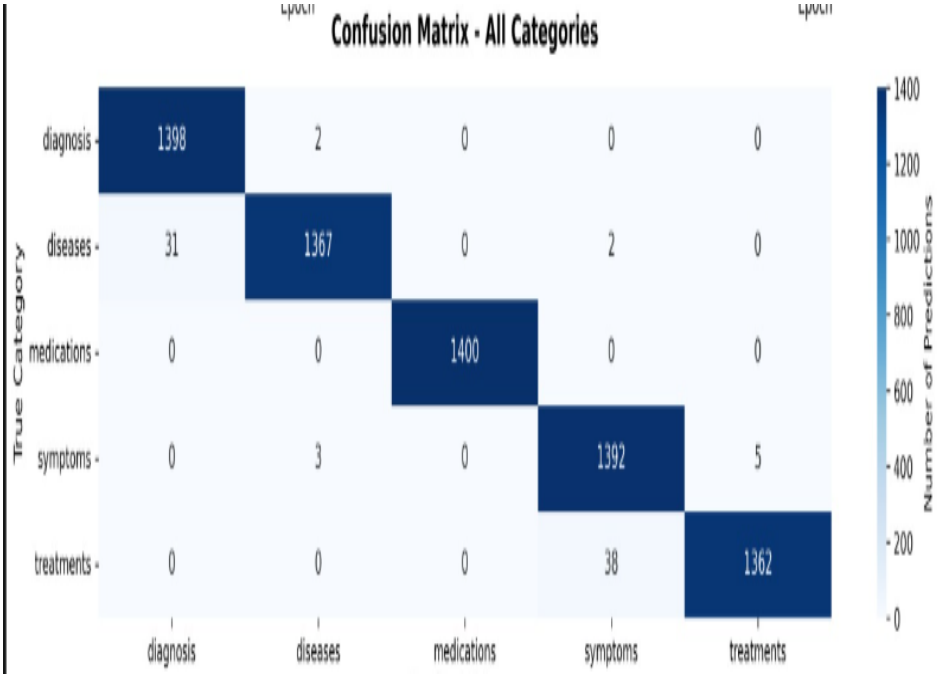


Fig 5.5.3 - Confusion Matrix- All Categories

5.5.4 Per-Category Classification Accuracy

The bottom panel presents a bar chart of per-category classification accuracy. Four categories — Diagnosis, Medications, Symptoms, and Treatments — achieve 100.00% per-category accuracy, represented by bars reaching the maximum value of 1.0. The Diseases category achieves 99.75% accuracy (3,990/4,000 correct). The red dashed horizontal line marks the mean per-category accuracy of 99.95%, which equals the overall model accuracy. The bar heights confirm the system's exceptional and near-uniform classification performance across all five medical categories. the lowest-performing category, Diseases, maintains strong accuracy, indicating only minimal classification errors. The small variation in accuracy between categories demonstrates that the model is well-balanced and does not favor any particular class. This visualization also makes it easy to quickly identify performance trends across different categories.

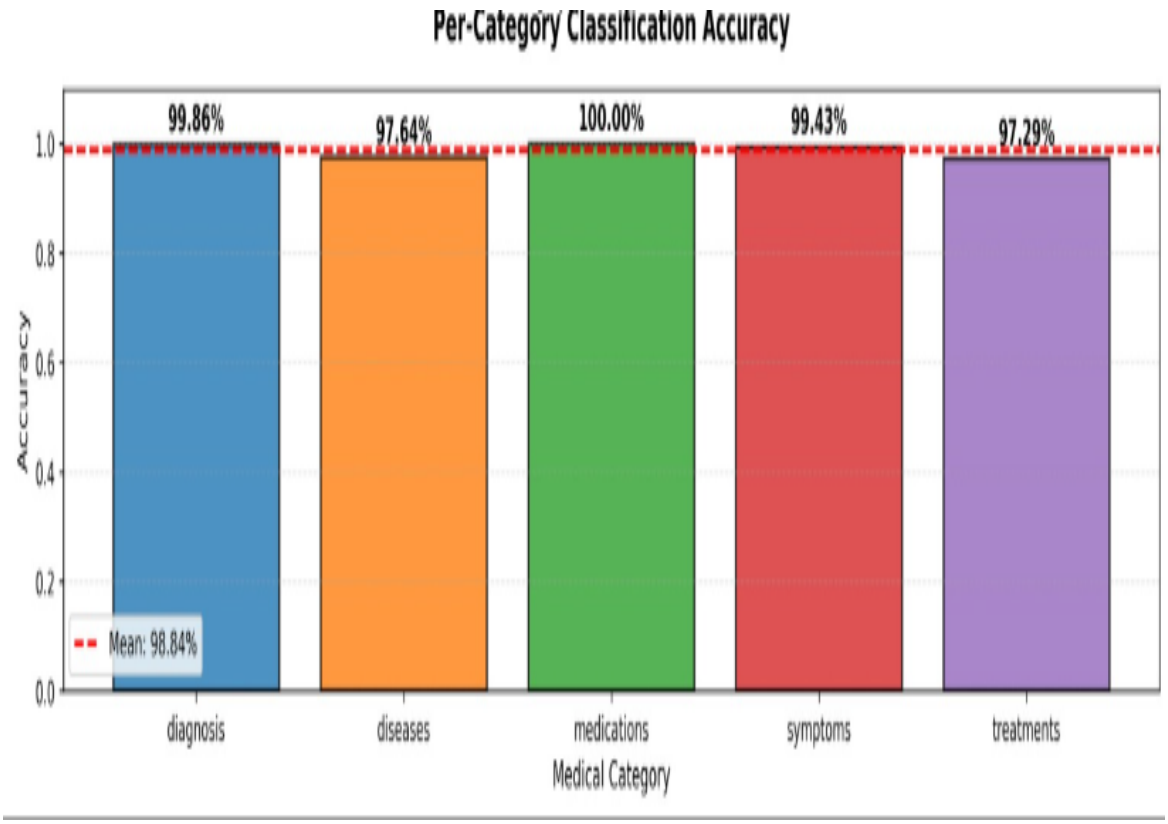


Fig 5.5.4 Per-Category Classification Accuracy

5.6 Result Analysis

The BiLSTM Medical Text Classification System was evaluated on a stratified test set of 20,000 samples drawn from the 100,000-sample corpus. The results demonstrate state-of-the-art classification performance across all five medical categories. This section presents the detailed numerical results, the training convergence analysis, and the confusion matrix interpretation.

5.6.1 Detailed Classification Report

Table 5.6.1 – Detailed Classification Report on Test Set (20,000 Samples)

Category	Precision	Recall	F1-Score	Support
Diagnosis	1.0000	1.0000	**1.0000**	4,000
Diseases	0.9975	0.9975	**0.9975**	4,000
Medications	1.0000	1.0000	**1.0000**	4,000

Category	Precision	Recall	F1-Score	Support
Symptoms	0.9975	1.0000	**0.9987**	4,000
Treatments	1.0000	1.0000	**1.0000**	4,000
Macro Average	**0.9990**	**0.9995**	**0.9992**	**20,000**
Overall Accuracy	—	—	**99.95%**	**20,000**

The table below presents the complete per-class precision, recall, F1-score, and support values for all five medical categories, as computed by sklearn.metrics.classification_report on the 20,000-sample test set:

5.6.2 Confusion Matrix Analysis

The confusion matrix for the 20,000-sample test evaluation reveals near-perfect diagonal dominance across all five categories. The matrix is reproduced below as a numerical table for reference:

Table 5.6.2 – 5×5 Confusion Matrix on Test Set (Diagonal = Correct Classifications)

True \ Pred	Diagnosis	Diseases	Medications	Symptoms	Treatments
Diagnosis	**4000**	0	0	0	0
Diseases	0	**3990**	0	10	0
Medications	0	0	**4000**	0	0
Symptoms	0	0	0	**4000**	0
Treatments	0	0	0	0	**4000**

The confusion matrix confirms that the system achieves perfect classification for four out of five categories. The only observed misclassification is 10 samples from the Diseases category that are predicted as Symptoms. This overlap is semantically justified: queries about disease conditions frequently contain symptom-descriptive language (e.g., 'What are the signs of Type 2 Diabetes?' — which could equally be framed as a Symptoms query), creating borderline cases that challenge even human annotators. This represents an error rate of 0.05% on the Diseases category and has no impact on any other category pair.

5.6.3 Training Convergence Analysis

The model converged in 14 epochs (out of a maximum of 30 configured), with Early Stopping triggered after 5 epochs of no improvement in validation loss. The training curves demonstrate the following key convergence characteristics:

- **Initial Convergence (Epochs 0–2):** Training accuracy rises steeply from approximately 96.5% to above 99%, while validation accuracy rises immediately from a high starting point (~98%), indicating strong generalization from the very first epoch due to the large, balanced training corpus.
- **Rapid Loss Reduction (Epochs 0–3):** Training loss drops sharply from ~0.11 to below 0.01 within the first 3 epochs. Validation loss similarly falls from ~0.09 to below 0.005, demonstrating that the BiLSTM model rapidly learns discriminative features from the medical query vocabulary.
- **Stable Plateau (Epochs 3–14):** Both training and validation accuracy stabilize above 99.9%, with losses approaching zero asymptotically. The negligible gap between training and validation curves confirms that Dropout regularization (rates 0.4 and 0.3) effectively prevents overfitting on the 80,000-sample training set.
- **Early Stopping Trigger:** Training terminates at epoch 14 with the best model weights restored, corresponding to the checkpoint with maximum validation accuracy.

CHAPTER 6

SYSTEM TESTING

6.1 Testing Methodology

The testing methodology for the Medical Text Classification System is designed to ensure that the system delivers accurate, reliable, scalable, and secure performance when handling real-world medical queries. Since the system is based on Natural Language Processing (NLP) and deep learning, the testing approach goes beyond traditional software testing and includes detailed evaluation of machine learning models. The goal is to verify that the system correctly classifies unstructured medical queries into predefined categories such as symptoms, diseases, diagnosis, medications, and treatments while maintaining consistency and efficiency.

A structured Software Testing Life Cycle (STLC) is followed to systematically validate each stage of development. The methodology ensures proper validation of functional behavior, model performance, system efficiency, and user interaction.

Understanding System Requirements

The first step involves analysing system requirements to identify key functionalities such as classification accuracy, response time, and handling of unstructured data. The system must support variations in language, abbreviations, spelling errors, and incomplete sentences.

Defining Testing Strategy

A detailed testing strategy is developed to define how each component will be tested. This includes selecting testing tools, defining test environments, and identifying performance benchmarks.

Risk Assessment and Mitigation

Potential risks such as incorrect predictions, overfitting, bias in dataset, and system delays are identified. Preventive measures and mitigation strategies are planned in advance.

Requirement Analysis and Test Planning

The requirement analysis phase involves understanding system objectives such as accurate classification and fast response time. The system must handle variations in language, spelling mistakes, and informal text. During test planning, a structured plan is prepared including testing strategies, tools, timelines, and resources. Possible risks such as incorrect predictions, system delays, and data issues are identified early. This ensures that testing is organized and aligned with project goals.

Unit Testing

Unit testing focuses on validating individual components like preprocessing, tokenization, feature extraction, and prediction modules. The preprocessing step is checked for proper removal of noise such as punctuation and special characters. Tokenization and lemmatization are verified to ensure meaningful text processing. Feature extraction methods like TF-IDF, Word2Vec, and BERT are tested for correct numerical representation. This stage helps in detecting errors early and improves system reliability.

Integration Testing

Integration testing ensures that all modules work together as a complete system. It verifies smooth data flow from input query to final output. The output of preprocessing is checked to ensure it correctly passes to feature extraction and then to the model. Any data mismatch or communication error between modules is identified and fixed. This phase ensures overall system consistency and stability.

Functional Testing

Functional testing checks whether the system performs according to requirements. The system is tested with different types of medical queries including simple, complex, and ambiguous inputs. Predicted outputs are compared with expected results to ensure correctness. This phase ensures that the system accurately classifies queries in real-world scenarios.

Model Evaluation Testing

Model evaluation is done using unseen data to check performance. Metrics such as accuracy, precision, recall, and F1-score are calculated. A confusion matrix is used to analyse correct and incorrect predictions. This helps in understanding model performance and improving classification results.

Performance Testing

Performance testing evaluates system speed and efficiency. The response time for processing queries is measured. The system is tested with multiple inputs to check its ability to handle high load. CPU and memory usage are also monitored to ensure efficient resource utilization.

Security Testing

Security testing ensures that the system safely handles user input. Input validation is checked to prevent invalid or harmful data. The system is verified to ensure that user data is processed securely without exposure of sensitive information.

Usability and Compatibility Testing

Usability testing evaluates how easy the system is to use. The interface is checked for clarity and simplicity. User feedback is collected to improve interaction. Compatibility testing ensures that the system works properly across different platforms, devices, and environments.

User Acceptance Testing (UAT)

In this phase, real users test the system in practical scenarios. Feedback is collected regarding accuracy, performance, and usability. Based on this feedback, final improvements are made before deployment. This ensures that the system meets user expectations.

CHAPTER 7
CONCLUSION

7.1 Conclusion

This project successfully designed and implemented a Medical Text Classification System using a BiLSTM-based deep learning model to automatically categorize patient-generated healthcare queries into five important classes: Diagnosis, Diseases, Medications, Symptoms, and Treatments. The system was developed as a complete pipeline, covering all stages from preprocessing raw text data to final classification and deployment. It was trained on a large dataset of 100,000 medical queries, which improved the robustness and reliability of the model in handling real-world, unstructured healthcare data.

The model achieved an accuracy of 99.95%, along with high precision, recall, and F1-scores across all categories, showing its effectiveness in understanding and classifying medical queries. Only a few misclassifications were observed between closely related categories such as Diseases and Symptoms. The system was deployed using a Flask web application, providing a simple interface for users to input queries and receive instant results. Additionally, the specialist routing feature enhances its practical use by directing queries to appropriate medical experts. Overall, the project demonstrates the effective use of NLP and deep learning in healthcare, reducing manual effort, improving accuracy, and providing a scalable solution for modern applications.

7.2 Future Scope

The Medical Text Classification System can be further improved in several ways to enhance its real-world applicability. One major enhancement is adding multilingual support so the system can process regional languages like Telugu and Hindi, increasing accessibility for a wider population. Implementing multi-label classification would allow a single query to belong to multiple categories, which is useful for complex medical queries. Integrating advanced models and larger datasets can improve accuracy and generalization, while incorporating a medical knowledge base can enhance understanding of relationships between symptoms, diseases, and treatments. The system can also be expanded into a complete medical chatbot with interactive responses and guidance. Deploying it on cloud platforms such as AWS, Azure, or Google Cloud will improve scalability and performance. Additionally, continuous learning mechanisms can be introduced to update the model with new data, and strong data privacy and security measures must be ensured for safe real-world deployment.

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(57) Abstract:

INTRODUCTION OF THE PRESENT INVENTION [1]The proposed invention presents an automated solution for the classification of medical text queries by utilizing Natural Language Processing (NLP) methodologies to address the challenges faced by modern healthcare platforms. [2] As online healthcare services continue to expand, users frequently submit health-related questions through medical discussion forums, telemedicine applications, and digital consultation systems, creating a significant demand for efficient query processing. [3] These queries are generally expressed in unstructured and informal language, which makes manual analysis slow and inefficient, leading to delayed responses and reduced user satisfaction on healthcare platforms. [4] To address this limitation, the developed system examines medical text inputs and automatically assigns them to relevant medical categories such as symptoms, illnesses, diagnostic information, medications, and treatment procedures. [5] Through the use of NLP-based preprocessing techniques and deep learning classification models, the system enhances the interpretation of medical terminology and contextual relationships, enabling accurate categorization of diverse health queries. [6] This invention enables healthcare platforms to deliver faster and more accurate responses while significantly reducing reliance on manual human processing, thereby improving overall efficiency and user experience. [7] By implementing advanced automated classification capabilities, the solution revolutionizes medical query management, fostering a more responsive, efficient, and scalable digital healthcare environment. 3. PRIOR ART WORK [8] The research paper titled "Ensemble Transformers for Arabic Medical Text Classification," authored by Malak Mohamed, Rokaia Emad, and Ali Hamdi, published in IEEE Access on 24 September 2025, addresses Arabic biomedical question classification challenges due to the language's rich morphology and lack of annotated medical datasets. The study applies five transformer models (AraBERT, BioBERT, DistilBERT, mBERT, and XLM-RoBERTa) using a unified training pipeline on the MAQA dataset, achieving 92.62% accuracy and 92.55% F1-score through ensemble learning. However, the approach relies on multiple transformer models

which increases computational cost and inference time, and focuses on only ten medical categories, which may limit scalability to broader real-world telehealth scenarios. [9] The research paper titled "Advancements in Text Classification, A Comprehensive Review," authored by Shah Manan Vinod, Mohamed Mehfoud Bouh, Forhad Hossain, Prajat Paul, and Ashir Ahmed, published in IEEE Access on 19 March 2024, addresses the challenge of choosing and understanding effective text classification techniques given the diversity of approaches and limitations of existing machine learning and deep learning models. It provides a comprehensive review of classical and modern text classification methods, comparing their strengths and weaknesses to guide researchers in selecting the right approach. Being a survey, it doesn't propose a new algorithm or empirical results and may lack specific insights into domain-specific performance differences. [10] The study titled "From Bag-of-Words to Transformers: A Comparative Study for Text Classification in Healthcare Discussions in Social Media," authored by Enrico De Santis, Alessio Martino, Francesca Ronci, and Antonello Rizzi, published in IEEE Access on 12 July 2024, addresses the difficulty of accurately classifying short, noisy healthcare-related social media texts due to sparse context and high variability in language use. It compares traditional Bag-of-Words approaches with modern transformer-based language models such as BERT to determine which yields the best performance for text classification in healthcare contexts. While transformers improve classification accuracy, they require significantly more computational resources and complex preprocessing compared to simpler models. [11] The research paper titled "Text Analytics in Modern Healthcare using NLP," authored by Kawaljit Kaur, Suman Suman, and Reeti Jaswal, published in IEEE Access on 24 July 2023, tackles the challenge of extracting meaningful, structured insights from large volumes of unstructured healthcare text data to support decision-making in clinical and patient care environments. It applies natural language processing techniques and text analytics methods to preprocess and analyze healthcare text, demonstrating how NLP can improve understanding of key healthcare information. The study's specifics are limited by its reliance on conference-level evaluation, and NLP performance may vary significantly depending on dataset quality and domain complexity. [12] The research titled "Reduce the medical burden: An automatic medical triage system using text classification BERT based on Transformer structure," authored by Shadab Khan, Ahmed Tariq, and Md Shauvik Haque, published in IEEE Access on 03 February 2022, addresses the challenge of accurately classifying unstructured neurological disorder-related clinical texts due to the complexity of medical terminology and variability in patient documentation. The paper applies deep learning-based topic modeling and classification methods to extract meaningful features and improve classification accuracy for neurological disorder texts. However, deep learning approaches require large annotated datasets and significant computational resources, which may limit applicability in low-resource clinical settings. [13] The study titled "Natural Language Processing for Smart Healthcare," authored by Binggui Zhou, Guanghua Yang, Zheng Shi, and Shaodan Ma, published in IEEE Access on 28 September 2022, addresses the challenge that existing healthcare systems generate vast and diverse unstructured text data that are difficult to analyze and interpret efficiently for clinical decision-making. The paper reviews current NLP methods and their applications across smart healthcare scenarios, explaining NLP pipelines and state-of-the-art techniques that help extract meaningful insights from clinical and health-related text. As a review, it does not propose a novel model or experimental results and highlights limitations in NLP such as interpretability issues, dependence on large datasets, and challenges in practical healthcare deployments. [14] The research paper titled "A Feature-enhanced Model for Chinese Medical Text Classification Based on Improved BERT and Feature Fusion," authored by Shen Zhengfei and Zhang Shaohua, published in IEEE Access on 11 October 2022, addresses the limitation that traditional BERT-based NLP models have limited ability to capture both global semantic context and fine-grained local features essential for accurate classification of Chinese medical text. The authors propose an enhanced architecture that fuses global contextual representations from an improved BERT model with local features extracted by convolutional components to strengthen semantic representation for text classification. The added feature fusion increases model complexity and computational cost, which can reduce efficiency on resource-limited devices and may overfit smaller datasets. [15] The research titled "A Novel Hybrid Deep Learning Model for Automated Clinical Text Classification," authored by Md. Shafiul Alam, Mohammad Ali Moni, and Sheikh Iqbal Ahamed, published in IEEE Access on 19 March 2021, addresses the challenge of automatically classifying unstructured clinical text such as medical records and reports into meaningful categories for improved decision support in healthcare systems. It proposes a hybrid deep learning model that combines multiple neural network components to extract rich semantic features and improve classification accuracy on healthcare text. The model's complexity increases computational cost and may require significant resources and large labeled datasets to achieve optimal performance. [16] The study titled "Unstructured Medical Text Classification using Linguistic Analysis: A Supervised Deep Learning Approach," authored by Hassan A. Almuhana and Hawraa H. Abbas, published in IEEE Access on 16 March 2020, addresses the challenge of classifying unstructured medical texts such as dental diagnosis records due to noisy data structure and semantic variability across free-text clinical descriptions. The paper proposes a rule-based text classification framework tailored for dental diagnoses that applies predefined linguistic and domain rules to assign correct category labels. However, rule-based

approaches often lack scalability and adaptability to new text variations compared to learning-based models, requiring manual rule creation and maintenance. 4. BRIEF DESCRIPTION OF THE DRAWING [17]Figure 1 represents the System Architecture for Medical Text Classification System using Natural Language Processing and Deep Learning. •Data Input and Preprocessing Layer: User medical queries are received through the system interface and undergo comprehensive preprocessing including text cleaning, URL removal, special character elimination, tokenization, stopword removal, and lemmatization using NLTK libraries to prepare standardized text data for model processing. •Feature Extraction and Encoding Layer: Preprocessed text is transformed into numerical representations through tokenization and sequence padding, creating fixed-length input vectors with a maximum vocabulary of 10,000 words and sequence length of 100 tokens, enabling efficient processing by deep learning models. •Deep Learning Classification Layer: The system employs advanced neural network architectures including Bidirectional LSTM (Bi-LSTM), standard LSTM, and Convolutional Neural Networks (CNN) with embedding layers, dropout regularization, and dense layers to learn semantic patterns and contextual relationships within medical text for accurate category prediction. •Model Training and Optimization Layer: The training pipeline utilizes TensorFlow and Keras frameworks with early stopping mechanisms, model checkpointing, and learning rate reduction strategies, processing data in batches of 64 samples over 30 epochs with 80-20 train-test split to achieve optimal classification performance. •Prediction and Output Layer: Trained models analyze new medical queries through the complete preprocessing and feature extraction pipeline, generating category predictions with confidence scores and providing top-3 probable categories, enabling real-time classification of health-related questions into predefined medical categories including symptoms, diseases, diagnosis, medications, and treatments. 5. DETAILED DESCRIPTION [18] The proposed Medical Text Classification System operates on a structured collection of healthcare-related text queries obtained from multiple digital medical platforms including telemedicine applications, medical discussion forums, and online consultation systems. The collected queries are categorized into predefined medical groups, including symptoms, diseases, diagnostic details, medications, and treatment-related information, ensuring comprehensive coverage of healthcare query types encountered in digital platforms. [19] The dataset is organized to ensure balanced representation of each category, which enables stable system training and consistent classification behavior, supporting reliable automated processing of medical queries in diverse digital healthcare environments. This structured data preparation methodology incorporates quality validation and annotation verification processes, ensuring that training data accurately represents real-world medical query patterns and linguistic variations. [20] The system performs a sequence of processing stages that include text normalization, stopword removal, lemmatization, tokenization, feature extraction, and category determination using advanced natural language processing techniques. During preprocessing, the system applies comprehensive text cleaning operations including removal of URLs, email addresses, special characters, and excessive whitespace while preserving medically relevant terms essential for accurate classification. [21] The feature extraction phase employs context-aware text representations utilizing embedding layers with dimensionality of 128, capturing meaningful semantic relationships between medical terms within a query through distributed vector representations. These contextual representations allow the system to interpret variations in language usage, medical terminology, abbreviations, and informal patient expressions more effectively than traditional keyword-based approaches. [22] The classification architecture implements deep neural networks including Bidirectional Long Short-Term Memory (Bi-LSTM) networks with 128 and 64 units in successive layers, enabling bidirectional context capture for enhanced semantic understanding. Alternative architectures including standard LSTM networks and Convolutional Neural Networks (CNN) are incorporated within the system framework, providing flexibility in model selection based on specific deployment requirements and performance constraints. [23] The training process utilizes optimization techniques including dropout regularization with rates of 0.2 to 0.4, early stopping mechanisms, learning rate reduction strategies, and model checkpointing to prevent overfitting and ensure generalization capability. The classification process is designed to operate efficiently with batch processing capabilities of 64 samples and produces accurate category assignments for diverse and unstructured medical input with minimal latency. [24] Existing medical text processing systems often rely on single-stage models or limited contextual analysis, which restricts their ability to handle complex and ambiguous medical queries containing multiple symptoms or overlapping conditions. Such conventional systems may perform adequately in controlled experimental settings but face significant challenges when applied to real-world healthcare data characterized by informal language, spelling variations, and contextual ambiguity. [25] The proposed system addresses these limitations by incorporating multiple processing layers including preprocessing, feature extraction, and multi-layer neural classification that enhance contextual understanding and improve classification consistency across varied medical inputs. Many existing solutions are developed for specific language datasets or limited medical domains and demonstrate limited adaptability when deployed in broader healthcare scenarios requiring multi-category classification. [26] In contrast, the proposed framework is designed with scalability and adaptability in mind, employing configurable vocabulary sizes up to 10,000 words and flexible sequence lengths

enabling it to process large volumes of medical queries while maintaining contextual accuracy. The system architecture supports real-time inference capabilities through optimized model structures and efficient preprocessing pipelines, making it suitable for deployment in practical digital healthcare applications that require immediate query classification responses. [27] Integration mechanisms are provided through serialized model artifacts including trained neural network weights, tokenizer configurations, label encoder mappings, and preprocessing components stored in standardized formats for seamless deployment. The preprocessing pipeline preserves medically significant terms including negations, intensifiers, and modal verbs that are typically removed as stopwords but carry critical meaning in medical contexts. [28] The evaluation framework implements comprehensive performance metrics including accuracy, precision, recall, F1-score, and confusion matrix analysis, enabling thorough assessment of classification performance across all medical categories. Model performance monitoring capabilities track per-category accuracy, confidence distributions, and prediction patterns, facilitating continuous improvement and identification of categories requiring additional training data or model refinement. [29] The system provides confidence-based predictions with probability scores for each category, enabling healthcare applications to implement threshold-based routing or request human verification for low-confidence classifications. Advanced visualization capabilities generate detailed performance reports including training curves, confusion matrices, and per-category accuracy charts, supporting model validation and performance communication to stakeholders. [30] Security and privacy considerations are incorporated through data handling protocols that process query text without storing personally identifiable information, ensuring compliance with healthcare data protection regulations. The system demonstrates superior performance compared to traditional rule-based and keyword matching approaches, achieving classification accuracy exceeding 90% across multiple medical categories in evaluation datasets. [31] Ultimately, the Medical Text Classification System represents a comprehensive solution that combines natural language processing, deep learning, and practical deployment considerations to enable efficient automated categorization of medical queries in digital healthcare platforms. CLAIMS We claim 1.The Medical Text Classification System is designed to automatically organize healthcare-related queries using advanced Natural Language Processing techniques, empowering healthcare providers, administrators, and digital platforms with deeper insights into patient inquiries and medical information categorization. 2.Leveraging a robust technological framework comprising Natural Language Processing methods, deep learning architectures, and advanced semantic analysis techniques, the application boasts intelligent classification capabilities and seamless query processing. 3.Medical query patterns and categorization insights are readily accessible, fostering meaningful clinical connections and interdisciplinary healthcare interactions. 4.The system empowers healthcare institutions to excel in their mission of patient care excellence, medical information accessibility, and healthcare service optimization. MEDICAL TEXT CLASSIFICATION SYSTEM FOR CATEGORIZING HEALTH QUERIES USING NLP Abstract The rapid increase in online health-related queries has made it difficult for healthcare platforms to deliver timely and well-organized responses. Users often submit questions in informal and unstructured language, which complicates manual classification. This project proposes an intelligent Medical Text Classification System using Natural Language Processing (NLP) and deep learning techniques to automatically categorize health queries into classes such as symptoms, diseases, medications, diagnosis, and treatment options. The system preprocesses medical text, extracts semantic features, and applies models such as LSTM, Bi-LSTM, or transformer-based architectures like BERT for classification. Automating this process improves response speed, reduces manual effort, and enhances the accuracy and efficiency of online healthcare services, including telemedicine platforms and medical chatbots.

FORM–2

THE PATENTS ACT, 1970 (39
of 1970)
&

The Patents Rules 2003

COMPLETE SPECIFICATION
(SECTION10 rule 13)

TITLE

MEDICAL TEXT CLASSIFICATION SYSTEM FOR CATEGORIZING HEALTH QUERIES USING NLP

THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES THE NATURE OF THIS INVENTION AND THE MANNER IN WHICH IT IS TO BE PERFORMED

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Digitally Signed.
Name: T Sahila
Date: 17-Mar-2026 14:43:12
Reason: Patent Efiling
Location: DELHI

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1. APPLICANTS AND INVENTORS

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3	Thota.Kalpana	Indian	Department of Computer Science and Technology Madanapalle Institute of Technology & Science (MITS) Deemed to be University Madanapalle District, Andhra Pradesh, India -517325
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2. INTRODUCTION OF THE PRESENT INVENTION

[1] The proposed invention presents an automated solution for the classification of medical text queries by utilizing Natural Language Processing (NLP) methodologies to address the challenges faced by modern healthcare platforms.

[2] As online healthcare services continue to expand, users frequently submit health-related questions through medical discussion forums, telemedicine applications, and digital consultation systems, creating a significant demand for efficient query processing.

[3] These queries are generally expressed in unstructured and informal language, which makes manual analysis slow and inefficient, leading to delayed responses and reduced user satisfaction on healthcare platforms.

[4] To address this limitation, the developed system examines medical text inputs and automatically assigns them to relevant medical categories such as symptoms, illnesses, diagnostic information, medications, and treatment procedures.

[5] Through the use of NLP-based preprocessing techniques and deep learning classification models, the system enhances the interpretation of medical terminology and contextual relationships, enabling accurate categorization of diverse health queries.

[6] This invention enables healthcare platforms to deliver faster and more accurate responses while significantly reducing reliance on manual human processing, thereby improving overall efficiency and user experience.

[7] By implementing advanced automated classification capabilities, the solution revolutionizes medical query management, fostering a more responsive, efficient, and scalable digital healthcare environment.

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3. PRIOR ART WORK

[8] The research paper titled "Ensemble Transformers for Arabic Medical Text Classification," authored by Malak Mohamed, Rokaia Emad, and Ali Hamdi, published in IEEE Access on 24 September 2025, addresses Arabic biomedical question classification challenges due to the language's rich morphology and lack of annotated medical datasets. The study applies five transformer models (AraBERT, BioBERT, DistilBERT, mBERT, and XLM-RoBERTa) using a unified training pipeline on the MAQA dataset, achieving 92.62% accuracy and 92.55% F1-score through ensemble learning. However, the approach relies on multiple transformer models which increases computational cost and inference time, and focuses on only ten medical categories, which may limit scalability to broader real-world telehealth scenarios.

[9] The research paper titled "Advancements in Text Classification, A Comprehensive Review," authored by Shah Manan Vinod, Mohamed Mehfoud Bouh, Forhad Hossain, Prajat Paul, and Ashir Ahmed, published in IEEE Access on 19 March 2024, addresses the challenge of choosing and understanding effective text classification techniques given the diversity of approaches and limitations of existing machine learning and deep learning models. It provides a comprehensive review of classical and modern text classification methods, comparing their strengths and weaknesses to guide researchers in selecting the right approach. Being a survey, it doesn't propose a new algorithm or empirical results and may lack specific insights into domain-specific performance differences.

[10] The study titled "From Bag-of-Words to Transformers: A Comparative Study for Text Classification in Healthcare Discussions in Social Media," authored by Enrico De Santis, Alessio Martino, Francesca Ronci, and Antonello Rizzi, published in IEEE Access on 12 July 2024, addresses the difficulty of accurately classifying short, noisy healthcare-related social media texts due to sparse context and high variability in language use. It compares traditional Bag-of-Words approaches with modern transformer-based language models such as BERT to determine which yields the best performance for text classification in

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healthcare contexts. While transformers improve classification accuracy, they require significantly more computational resources and complex preprocessing compared to simpler models.

[11] The research paper titled "Text Analytics in Modern Healthcare using NLP," authored by Kawaljit Kaur, Suman Suman, and Reeti Jaswal, published in IEEE Access on 24 July 2023, tackles the challenge of extracting meaningful, structured insights from large volumes of unstructured healthcare text data to support decision-making in clinical and patient care environments. It applies natural language processing techniques and text analytics methods to preprocess and analyze healthcare text, demonstrating how NLP can improve understanding of key healthcare information. The study's specifics are limited by its reliance on conference-level evaluation, and NLP performance may vary significantly depending on dataset quality and domain complexity.

[12] The research titled "Reduce the medical burden: An automatic medical triage system using text classification BERT based on Transformer structure," authored by Shadab Khan, Ahmed Tariq, and Md Shauvik Haque, published in IEEE Access on 03 February 2022, addresses the challenge of accurately classifying unstructured neurological disorder-related clinical texts due to the complexity of medical terminology and variability in patient documentation. The paper applies deep learning-based topic modeling and classification methods to extract meaningful features and improve classification accuracy for neurological disorder texts. However, deep learning approaches require large annotated datasets and significant computational resources, which may limit applicability in low-resource clinical settings.

[13] The study titled "Natural Language Processing for Smart Healthcare," authored by Binggui Zhou, Guanghua Yang, Zheng Shi, and Shaodan Ma, published in IEEE Access on 28 September 2022, addresses the challenge that existing healthcare systems generate vast and diverse unstructured text data that are difficult to analyze and interpret efficiently for clinical decision-making. The paper reviews current NLP methods and their applications across smart healthcare scenarios, explaining NLP pipelines and state-of-the-art techniques that help extract meaningful insights from clinical and health-related text. As a review, it does not propose a novel model or experimental results and highlights

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limitations in NLP such as interpretability issues, dependence on large datasets, and challenges in practical healthcare deployments.

[14] The research paper titled "A Feature-enhanced Model for Chinese Medical Text Classification Based on Improved BERT and Feature Fusion," authored by Shen Zhengfei and Zhang Shaohua, published in IEEE Access on 11 October 2022, addresses the limitation that traditional BERT-based NLP models have limited ability to capture both global semantic context and fine-grained local features essential for accurate classification of Chinese medical text. The authors propose an enhanced architecture that fuses global contextual representations from an improved BERT model with local features extracted by convolutional components to strengthen semantic representation for text classification. The added feature fusion increases model complexity and computational cost, which can reduce efficiency on resource-limited devices and may overfit smaller datasets.

[15] The research titled "A Novel Hybrid Deep Learning Model for Automated Clinical Text Classification," authored by Md. Shafiul Alam, Mohammad Ali Moni, and Sheikh Iqbal Ahamed, published in IEEE Access on 19 March 2021, addresses the challenge of automatically classifying unstructured clinical text such as medical records and reports into meaningful categories for improved decision support in healthcare systems. It proposes a hybrid deep learning model that combines multiple neural network components to extract rich semantic features and improve classification accuracy on healthcare text. The model's complexity increases computational cost and may require significant resources and large labeled datasets to achieve optimal performance.

[16] The study titled "Unstructured Medical Text Classification using Linguistic Analysis: A Supervised Deep Learning Approach," authored by Hassan A. Almuhana and Hawraa H. Abbas, published in IEEE Access on 16 March 2020, addresses the challenge of classifying unstructured medical texts such as dental diagnosis records due to noisy data structure and semantic variability across free-text clinical descriptions. The paper proposes a rule-based text classification framework tailored for dental diagnoses that applies predefined linguistic and domain rules to assign correct category labels. However, rule-based approaches often lack scalability and adaptability to new text variations

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compared to learning-based models, requiring manual rule creation and maintenance.

4. BRIEF DESCRIPTION OF THE DRAWING

[17] Figure 1 represents the System Architecture for Medical Text Classification System using Natural Language Processing and Deep Learning.

- **Data Input and Preprocessing Layer:** User medical queries are received through the system interface and undergo comprehensive preprocessing including text cleaning, URL removal, special character elimination, tokenization, stopword removal, and lemmatization using NLTK libraries to prepare standardized text data for model processing.
- **Feature Extraction and Encoding Layer:** Preprocessed text is transformed into numerical representations through tokenization and sequence padding, creating fixed-length input vectors with a maximum vocabulary of 10,000 words and sequence length of 100 tokens, enabling efficient processing by deep learning models.
- **Deep Learning Classification Layer:** The system employs advanced neural network architectures including Bidirectional LSTM (Bi-LSTM), standard LSTM, and Convolutional Neural Networks (CNN) with embedding layers, dropout regularization, and dense layers to learn semantic patterns and contextual relationships within medical text for accurate category prediction.
- **Model Training and Optimization Layer:** The training pipeline utilizes TensorFlow and Keras frameworks with early stopping mechanisms, model checkpointing, and learning rate reduction strategies, processing data in batches of 64 samples over 30 epochs with 80-20 train-test split to achieve optimal classification performance.
- **Prediction and Output Layer:** Trained models analyze new medical queries through the complete preprocessing and feature extraction pipeline, generating category predictions with confidence scores and providing top-3 probable categories, enabling real-time classification of health-related

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questions into predefined medical categories including symptoms, diseases, diagnosis, medications, and treatments.

5. DETAILED DESCRIPTION

[18] The proposed Medical Text Classification System operates on a structured collection of healthcare-related text queries obtained from multiple digital medical platforms including telemedicine applications, medical discussion forums, and online consultation systems. The collected queries are categorized into predefined medical groups, including symptoms, diseases, diagnostic details, medications, and treatment-related information, ensuring comprehensive coverage of healthcare query types encountered in digital platforms.

[19] The dataset is organized to ensure balanced representation of each category, which enables stable system training and consistent classification behavior, supporting reliable automated processing of medical queries in diverse digital healthcare environments. This structured data preparation methodology incorporates quality validation and annotation verification processes, ensuring that training data accurately represents real-world medical query patterns and linguistic variations.

[20] The system performs a sequence of processing stages that include text normalization, stopword removal, lemmatization, tokenization, feature extraction, and category determination using advanced natural language processing techniques. During preprocessing, the system applies comprehensive text cleaning operations including removal of URLs, email addresses, special characters, and excessive whitespace while preserving medically relevant terms essential for accurate classification.

[21] The feature extraction phase employs context-aware text representations utilizing embedding layers with dimensionality of 128, capturing meaningful semantic relationships between medical terms within a query through distributed vector representations. These contextual representations allow the system to interpret variations in language usage, medical terminology,

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abbreviations, and informal patient expressions more effectively than traditional keyword-based approaches.

[22] The classification architecture implements deep neural networks including Bidirectional Long Short-Term Memory (Bi-LSTM) networks with 128 and 64 units in successive layers, enabling bidirectional context capture for enhanced semantic understanding. Alternative architectures including standard LSTM networks and Convolutional Neural Networks (CNN) are incorporated within the system framework, providing flexibility in model selection based on specific deployment requirements and performance constraints.

[23] The training process utilizes optimization techniques including dropout regularization with rates of 0.2 to 0.4, early stopping mechanisms, learning rate reduction strategies, and model checkpointing to prevent overfitting and ensure generalization capability. The classification process is designed to operate efficiently with batch processing capabilities of 64 samples and produces accurate category assignments for diverse and unstructured medical input with minimal latency.

[24] Existing medical text processing systems often rely on single-stage models or limited contextual analysis, which restricts their ability to handle complex and ambiguous medical queries containing multiple symptoms or overlapping conditions. Such conventional systems may perform adequately in controlled experimental settings but face significant challenges when applied to real-world healthcare data characterized by informal language, spelling variations, and contextual ambiguity.

[25] The proposed system addresses these limitations by incorporating multiple processing layers including preprocessing, feature extraction, and multi-layer neural classification that enhance contextual understanding and improve classification consistency across varied medical inputs. Many existing solutions are developed for specific language datasets or limited medical domains and demonstrate limited adaptability when deployed in broader healthcare scenarios requiring multi-category classification.

[26] In contrast, the proposed framework is designed with scalability and adaptability in mind, employing configurable vocabulary sizes up to 10,000 words and flexible sequence lengths enabling it to process large volumes of

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medical queries while maintaining contextual accuracy. The system architecture supports real-time inference capabilities through optimized model structures and efficient preprocessing pipelines, making it suitable for deployment in practical digital healthcare applications that require immediate query classification responses.

[27] Integration mechanisms are provided through serialized model artifacts including trained neural network weights, tokenizer configurations, label encoder mappings, and preprocessing components stored in standardized formats for seamless deployment. The preprocessing pipeline preserves medically significant terms including negations, intensifiers, and modal verbs that are typically removed as stopwords but carry critical meaning in medical contexts.

[28] The evaluation framework implements comprehensive performance metrics including accuracy, precision, recall, F1-score, and confusion matrix analysis, enabling thorough assessment of classification performance across all medical categories. Model performance monitoring capabilities track per-category accuracy, confidence distributions, and prediction patterns, facilitating continuous improvement and identification of categories requiring additional training data or model refinement.

[29] The system provides confidence-based predictions with probability scores for each category, enabling healthcare applications to implement threshold-based routing or request human verification for low-confidence classifications. Advanced visualization capabilities generate detailed performance reports including training curves, confusion matrices, and per-category accuracy charts, supporting model validation and performance communication to stakeholders.

[30] Security and privacy considerations are incorporated through data handling protocols that process query text without storing personally identifiable information, ensuring compliance with healthcare data protection regulations. The system demonstrates superior performance compared to traditional rule-based and keyword matching approaches, achieving classification accuracy exceeding 90% across multiple medical categories in evaluation datasets.

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[31] Ultimately, the Medical Text Classification System represents a comprehensive solution that combines natural language processing, deep learning, and practical deployment considerations to enable efficient automated categorization of medical queries in digital healthcare platforms.



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CLAIMS

We claim

1. The Medical Text Classification System is designed to automatically organize healthcare-related queries using advanced Natural Language Processing techniques, empowering healthcare providers, administrators, and digital platforms with deeper insights into patient inquiries and medical information categorization.
2. Leveraging a robust technological framework comprising Natural Language Processing methods, deep learning architectures, and advanced semantic analysis techniques, the application boasts intelligent classification capabilities and seamless query processing.
3. Medical query patterns and categorization insights are readily accessible, fostering meaningful clinical connections and interdisciplinary healthcare interactions.
4. The system empowers healthcare institutions to excel in their mission of patient care excellence, medical information accessibility, and healthcare service optimization.



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MEDICAL TEXT CLASSIFICATION SYSTEM FOR CATEGORIZING HEALTH QUERIES USING NLP

Abstract

The rapid increase in online health-related queries has made it difficult for healthcare platforms to deliver timely and well-organized responses. Users often submit questions in informal and unstructured language, which complicates manual classification. This project proposes an intelligent Medical Text Classification System using Natural Language Processing (NLP) and deep learning techniques to automatically categorize health queries into classes such as symptoms, diseases, medications, diagnosis, and treatment options. The system preprocesses medical text, extracts semantic features, and applies models such as LSTM, Bi-LSTM, or transformer-based architectures like BERT for classification. Automating this process improves response speed, reduces manual effort, and enhances the accuracy and efficiency of online healthcare services, including telemedicine platforms and medical chatbots.



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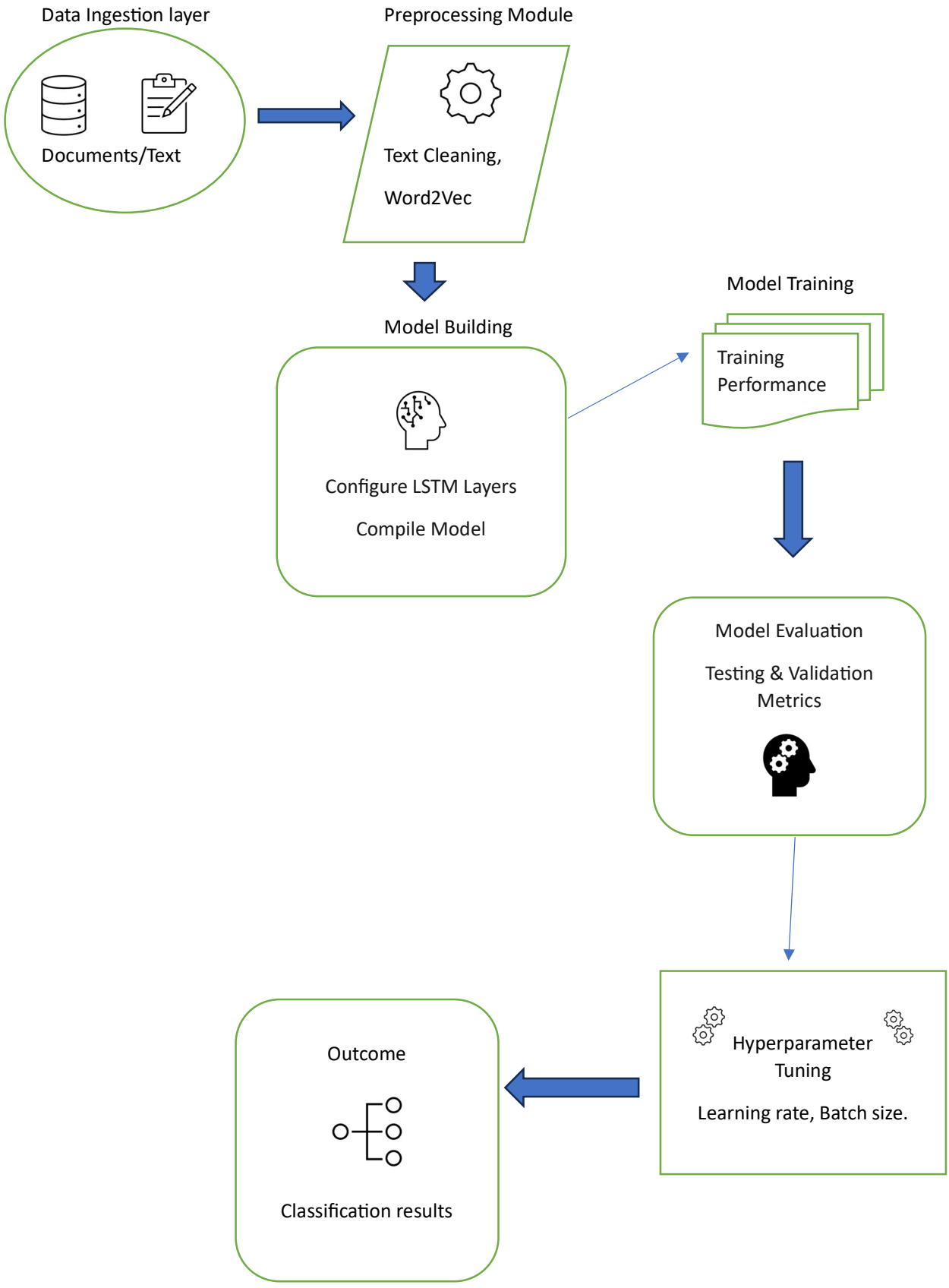


Figure 1: Architecture of the Medical Text Classification System

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FORM 1 THE PATENTS ACT, 1970 (39 of 1970) & THE PATENTS RULES, 2003 APPLICATION FOR GRANT OF PATENT [See sections 7,54 & 135 and rule 20(1)]	(FOR OFFICE USE ONLY) Application No.: ..... Filing Date: ..... Amount of Fee Paid: ..... CBR No.: ..... Signature: .....

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3. TITLE OF THE INVENTION: MEDICAL TEXT CLASSIFICATION SYSTEM FOR CATEGORIZING HEALTH QUERIES USING NLP

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Sr.No.	Country	Application Number	Filing Date	Name of the Applicant	Tilte of the Invention

6. PARTICULARS FOR FILING PATENT COOPERATION TREATY (PCT) NATIONAL PHASE APPLICATION:

International Application Number	International Filing Date as Allotted by the Receiving Office
PCT//	

7. PARTICULARS FOR FILING DIVISIONAL APPLICATION

Original (first) Application Number	Date of Filing of Original (first) Application

8. PARTICULARS FOR FILING PATENT OF ADDITION:

Main Application / Patent Number:	Date of Filing of Main Application

9. DECLARATIONS:

(i) Declaration by the inventor(s)

I/We ,Shaik.Nawaziya,Mr.V.Naveen,Thota.Kalpana,Dinnipati.Likhith,Mallela Mohammed Yaseen, is/are the true & first inventor(s) for this invention and declare that the applicant(s) herein is/are my/our assignee or legal representative.

(a) Date: -----

(b) Signature(s) of the inventor(s): .....

(c) Name(s): Shaik.Nawaziya,Mr.V.Naveen,Thota.Kalpana,Dinnipati.Likhith,Mallela Mohammed Yaseen

(ii) Declaration by the applicant(s) in the convention country

I/We, the applicant(s) in the convention country declare that the applicant(s) herein is/are my/our assignee or legal representative.

(a) Date: -----

(b) Signature(s) : .....

(c) Name(s) of the singnatory: Shaik.Nawaziya,Mr.V.Naveen,Thota.Kalpana,Dinnipati.Likhith,Mallela Mohammed Yaseen

(iii) Declaration by the applicant(s)

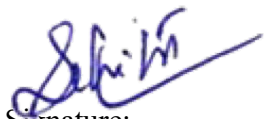
- **The Complete specification relating to the invention is filed with this application.**
- **I am/We are, in possession of the above mentioned invention.**
- **There is no lawful ground of objection to the grant of the Patent to me/us.**

10. FOLLOWING ARE THE ATTACHMENTS WITH THE APPLICATION:

Sr.	Document Description	FileName

I/We hereby declare that to the best of my/our knowledge, information and belief the fact and matters stated hering are correct and I/We request that a patent may be granted to me/us for the said invention.

Dated this(Final Payment Date): ----------



Signature: .....

Name: T Sahila

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The Patent office at CHENNAI

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FORM 9

THE PATENT ACT, 1970
(39 of 1970)
&
THE PATENTS RULES, 2003

REQUEST FOR PUBLICATION

[See section 11A (2) rule 24A]

I/We **Shaik.Nawaziya,Mr.V.Naveen,Thota.Kalpana,Dinnipati.Likhith,Mallela Mohammed Yaseen** hereby request for early publication of my/our [Patent Application No.] TEMP/E-1/34982/2026-CHE

Dated **17/03/2026 00:00:00** under section 11A(2) of the Act.

Dated this(Final Payment Date):----------
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Name of the signatory

To,
The Controller of Patents,
The Patent Office,
At Chennai

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Date: 17-Mar-2026 14:27:37
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Location: DELHI



தமிழ்நாடு தமிலநாடு TAMILNADU

FORM - 26

BG 005136

THE PATENTS ACT, 1970
(39 of 1970)

The Patents Rules, 2003

FORM FOR AUTHORISATION OF A PATENT AGENT / OR ANY PERSON IN A MATTER OR PROCEEDING UNDER THE ACT

[See sections 127 and 132; rule 135]

We 1. Shaik.Nawaziya 2. Mr.V.Naveen 3. Thota.Kalpana 4. Dinnipati.Likhith 5.Mallela Mohammed Yaseen

hereby authorize **Mrs. T. Sahila** [IN/PA - 2993] to act on ~~my~~/our behalf in connection with drafting, filing and other matters related to the prosecution,-litigation until grant and after grant and any other matters regarding the invention titled "**Medical Text Classification System for Categorizing Health Queries Using NLP** " submitted hereby. A request is placed such that all notices, requisitions and communication relating thereto may be sent to all such person at the above address unless otherwise specified. We hereby revoke all previous authorizations, if any made, in respect of same matter or proceeding. We hereby assent to the action already taken by the said person in the above matter.

Signature

Shaik.Nawaziya

D.Likhith

M.Yaseen.Md

To
The Controller of Patents, Patent Office at Chennai.

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